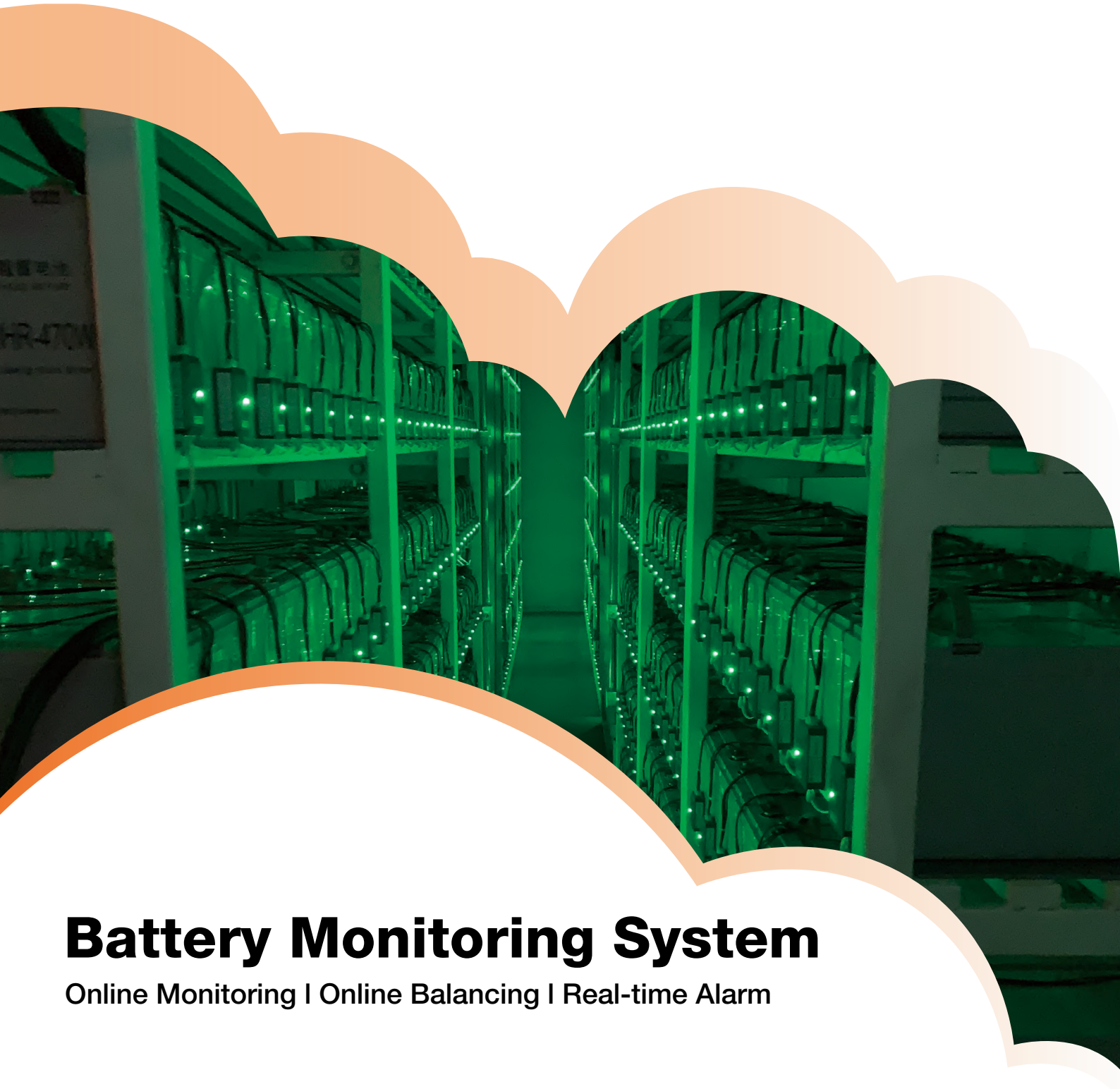
DFUN



Battery Monitoring System

Online Monitoring I Online Balancing I Real-time Alarm

DFUN (ZHUHAI) CO.,LTD.



Founded in April 2013 and based in Zhuhai, DFUN (ZHUHAI) CO., LTD. has established a strong presence across China, with regional offices in BEIJING, NANJING, and other cities, along with an overseas branch in DFUN Thailand. The company operates a modern 6,000 sqm production facility and has an annual capacity of over one million battery sensors, delivering high-quality solutions to global customers.

DFUN TECH emphasises quality and customer service. We offer hardware and software solutions for Battery Monitoring Systems, lithium battery packs, and battery capacity testers to valued clients worldwide.

OUR WORKSHOP








DFUN TECH

With an international professional sales network, DFUN products have been widely adopted in more than 80 countries and regions across the Asia-Pacific, Middle East, CIS, Europe, Africa, and South America. Our global presence includes subsidiaries such as the Thailand branch.

To ensure the quality of products,services,safety and environmental protection requirements,DFUN TECH implements the criteria of ISO9001,ISO14001,ISO45001,CE,FCC,Rohs,UL etc.

DFUN TECH will keep developing new solutions and offering the best options for our customers. We listen carefully to our partners and aim to provide a diverse, comprehensive, and effective BMS solution.


ISO9001


ISO14001


ISO45001



CE



ROHS



TTL



UL

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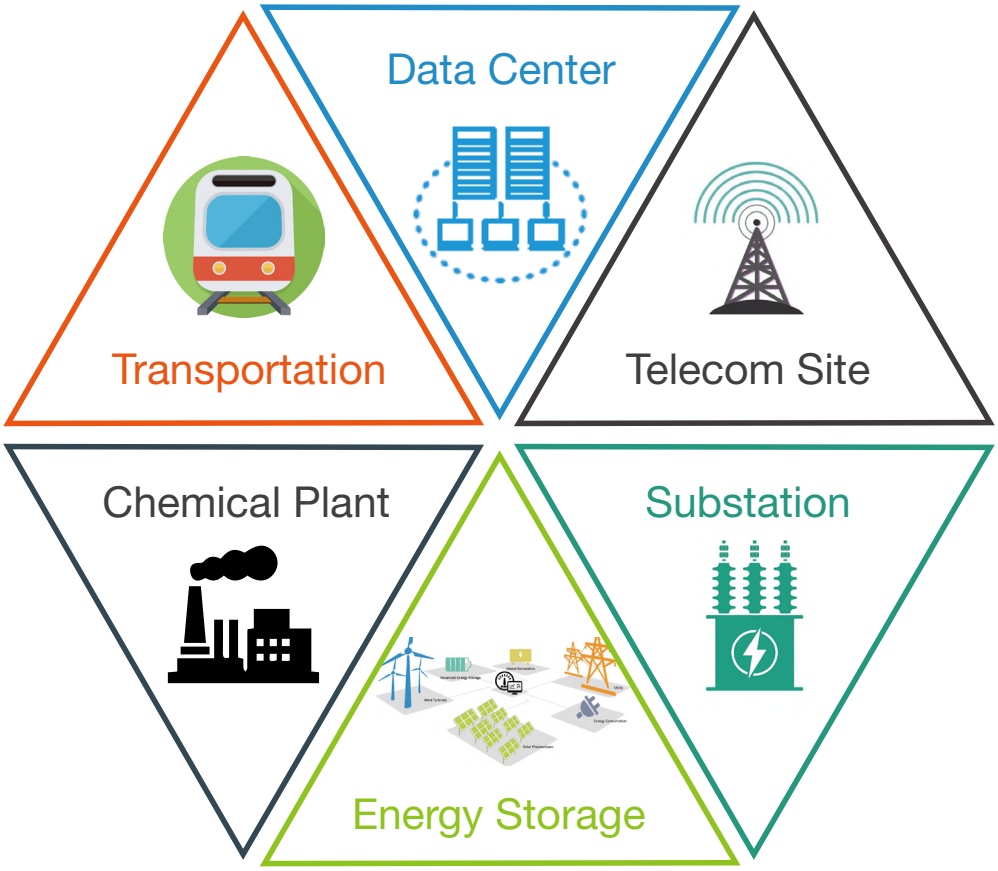
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Our Clients

Industry Application Solutions

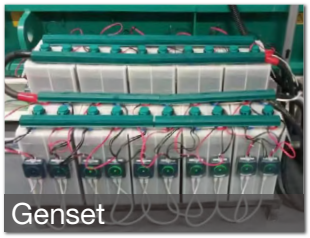



Data Center


Substation


Railway Transport


Hospital


Genset


Energy Storage


Telecom Site


Chemical Plant

Our Feature



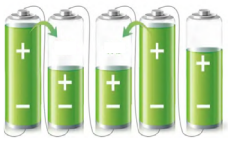
24/7 Online Monitoring

Continuously tracks battery voltage, charge/discharge current, internal resistance, temperature, SOC, SOH, and more. Provides accurate data reports with real-time alarms, enabling quick response to potential battery risks. Significantly reduces manual maintenance costs while ensuring safety and reliability.



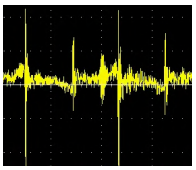
AI Data Analysis & Maintenance

The system leverages real-time big data and AI algorithms to transform raw battery data into valuable insights, delivering proactive risk warnings, lifecycle optimization, and strategic decision support.



Real-Time Intelligent Balancing

The system continuously analyzes uploaded battery data and automatically detects voltage imbalances. When imbalance occurs, it issues precise commands to ensure consistent voltage across all batteries, enhancing reliability, performance, and service life.



Ripple Voltage & Current Monitoring

The BMS precisely monitors ripple voltage and ripple current in real time. By detecting abnormal fluctuations caused by chargers, rectifiers, or power electronics, the system safeguards battery cells from hidden damage, reduces heat accumulation, and ensures long-term reliability.



Battery Early Warning

Through continuous monitoring and real-time modeling, the system identifies abnormal performance trends such as over-temperature, over-voltage, and capacity decline. Early detection enables predictive maintenance and minimizes downtime.



Breathing Alarm Light

The breathing light alarm system delivers real-time visual signals for voltage imbalance, temperature rise, or abnormal battery conditions. This intelligent design enhances situational awareness and ensures rapid response to potential faults.



Thermal Runaway Detection

Continuous monitoring of temperature gradients, internal resistance, and charging patterns allows the system to capture early signs of thermal runaway. With intelligent alarms and automatic intervention, the BMS minimizes fire hazards and protects assets.



Auto-Addressing & Time Saving

The BMS features intelligent autoaddressing technology, which automatically searches and configures device addresses without manual intervention. This plug-andplay capability simplifies installation, eliminates configuration errors, and significantly reduces commissioning time. By minimizing human workload, the system saves both labor costs and operational time,ensuring faster and more efficient deployment.

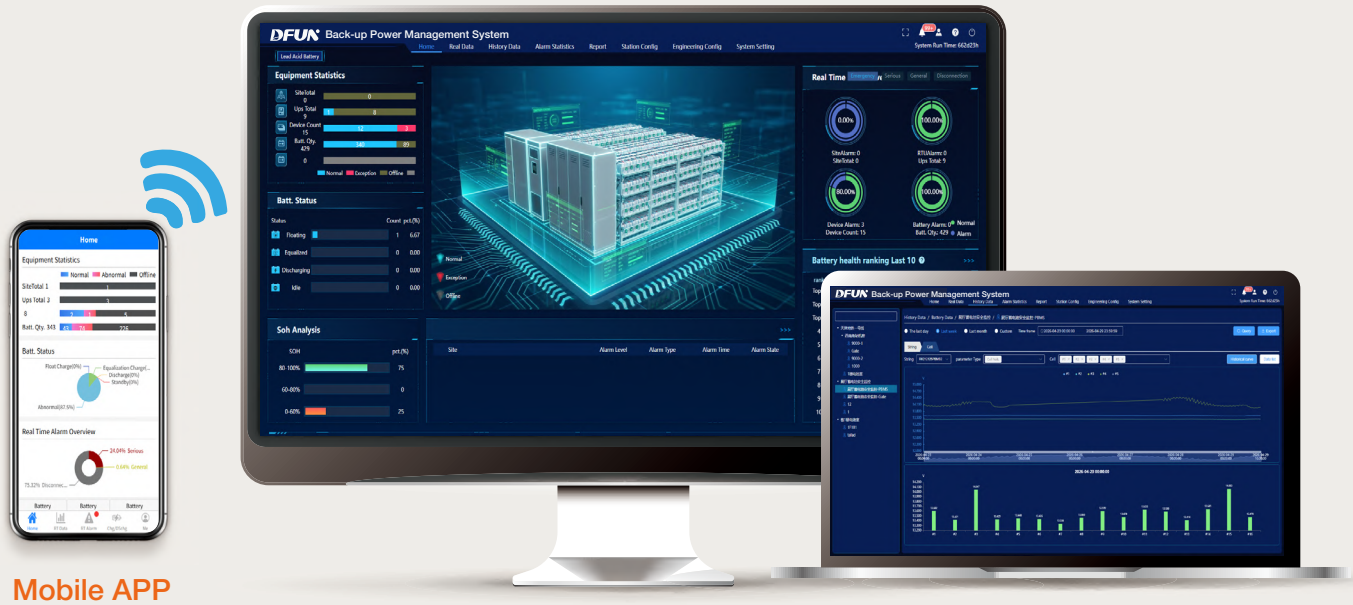


System Management Layer

Chapter 1

DFCS4200 Back-up Power Management System

DFCS4200 Back-up Power Management System



Mobile APP
SMS / Email Alarm

Unified Monitoring. Total Visibility.

Application

- Data center
- Telecommunications
- Substation & power utility
- Railway station & metro
- Bank
- Hospital
- Government
- Oil & Gas
- Factory
- Airport

...

Unified Multi-System Monitoring

UPS, BMS & Lithium Batteries in One Platform

- Integrate UPS, battery systems and energy assets
- Compatible with multiple brands and communication protocols
- One unified interface for all backup power systems

Cell-Level Monitoring & Early Warning

Detect Issues Before They Impact the System

- Real-time monitoring of voltage, temperature, resistance and current
- Identify battery imbalance and abnormal conditions earl
- Detect leakage, electrolyte level and environmental risks

Large-Scale Centralized Management

Monitor Up to 100,000 Batteries Across Sites

- Centralized monitoring across multiple locations
- Real-time system status and alarm overview
- Scalable for telecom, data center and utility applications

Intelligent Data Logging & AI Analysis

From Data to Predictive Maintenance

- Record historical battery performance and system logs
- Support manual capacity testing records and report generation
- AI-driven analysis for trend prediction and early fault detection
- Enable data-driven maintenance and lifecycle optimization

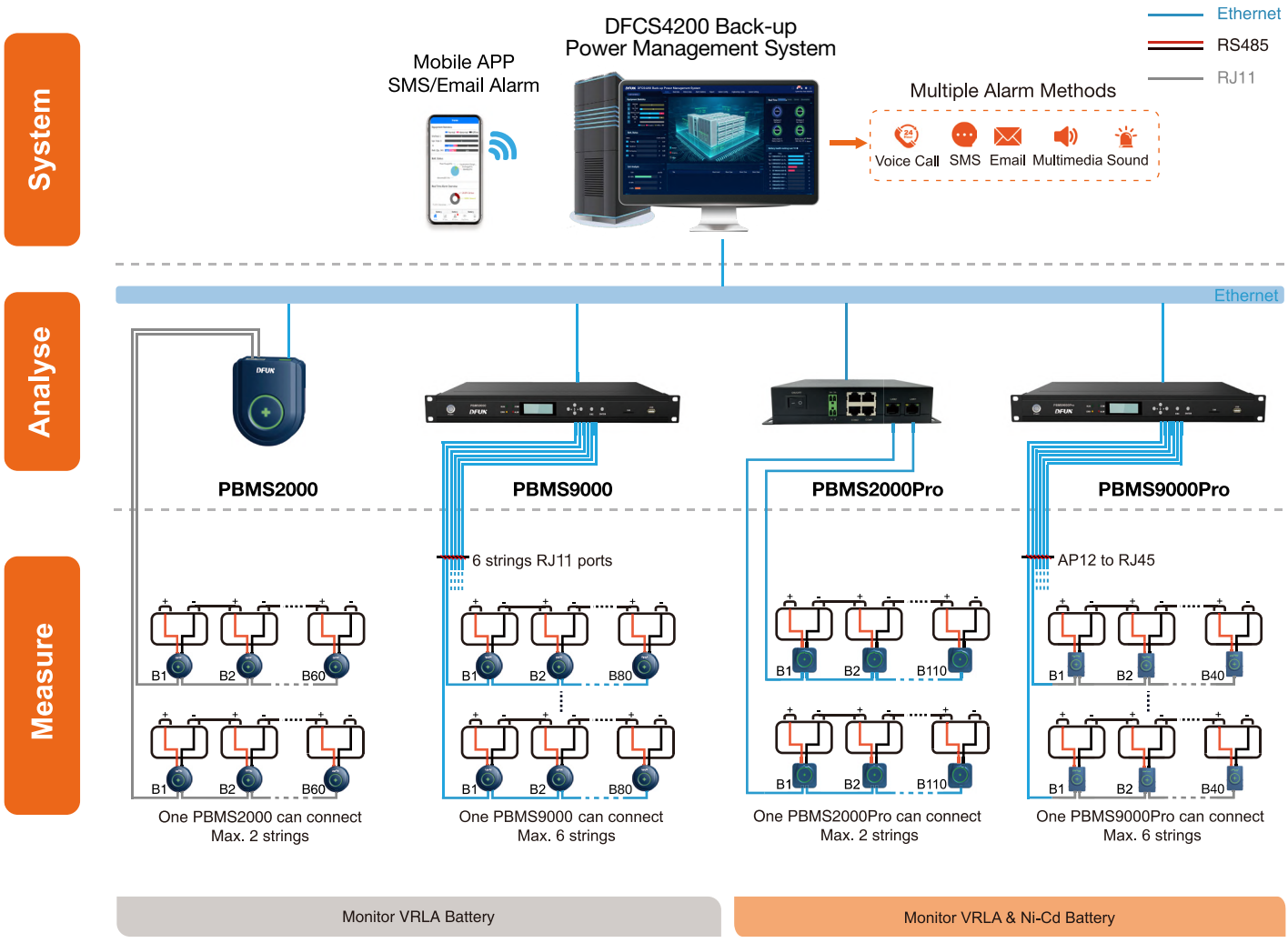
DFCS4200 Back-up Power Management System

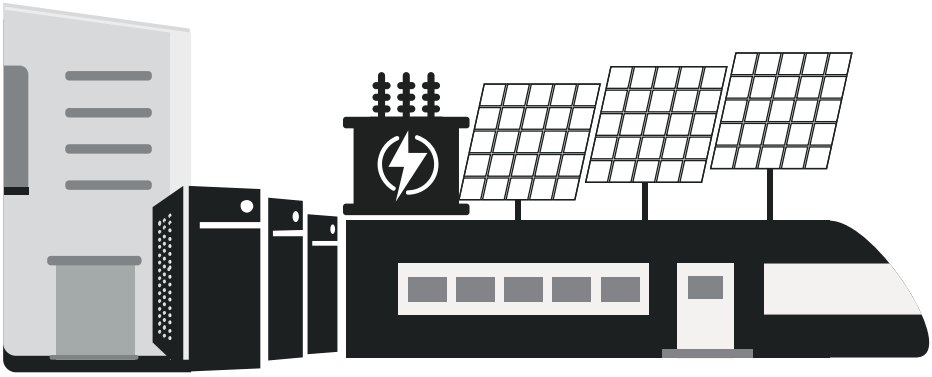
System Feature

- 24/7/365 real-time monitoring ≤100,000pcs batteries
- Supports cross-platform deployment (Windows, Linux)
- B/S structure with simple and UI set-up
- Realize all data display at one location, suitable for a project by large number of batteries or batteries located at different site
- Sent out timely alarm notifications for abnormal battery via SMS & Email alerts (support multiple contacts)
- Setpoint multi-level alarm
- Graphical and visual display of battery trends and faults
- Provides centralized data storage, data access and data management, export .csv format reports for data analysis
- Monitor ambient temperature & humidity
- Support monitoring thermal runaway



System Structure





BMS for Various Applications

Chapter 2

For Monitoring VRLA Battery

- PBMS2000 for Telecom Site
- PBMS9000 for Data Center

For Monitoring Ni-Cd / VRLA Battery

- PBMS2000Pro for Power and Rail Industries
- PBMS9000Pro for Substation / Chemical Plant

DFCS4200 Back-up
Power Management System



Mobile APP
SMS/Email Alarm



1 2 3 4 5



PBMS2000



PBMS2000Pro



PBMS9000



PBMS9000Pro



BTS Station



Substation



Data Center



Substation



Chemical Plant

PBMS2000
for Telecom Site

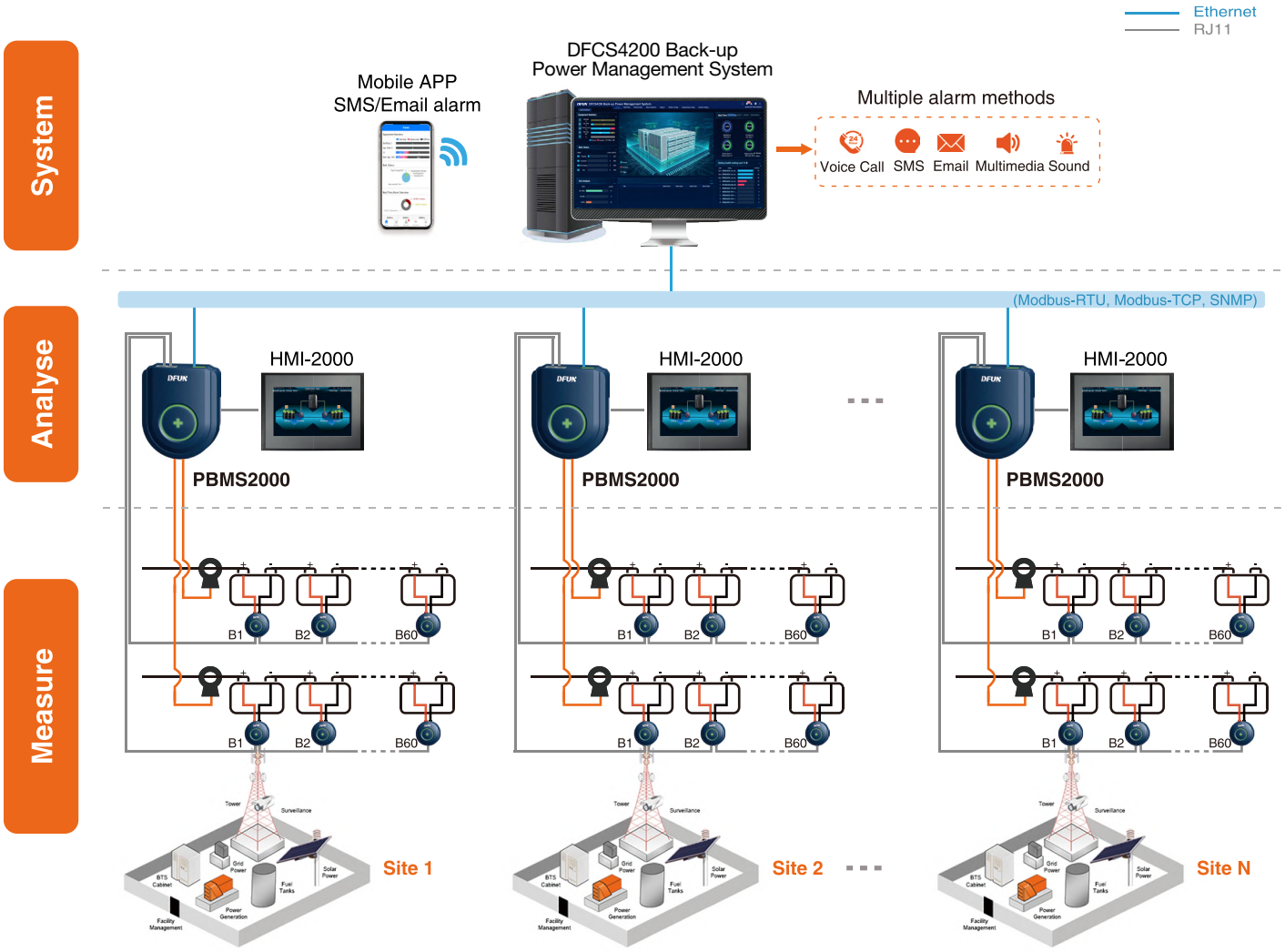


Feature

- Dedicated design for telecommunication 24V or -48V system
- 24/7hours on-line monitoring & remote alarm
- Cost competitive solution, support Max. 1 string, total 120pcs batteries,or Max. 2 strings, Max. 60pcs batteries each string
- Measure Lead-Acid battery
- Multi communication protocol (Modbus-RTU, MODBUS-TCP, SNMP)
- Comply with IEEE 1188-2005

PBMS2000 for Telecom Site

System Structure



Management Layer

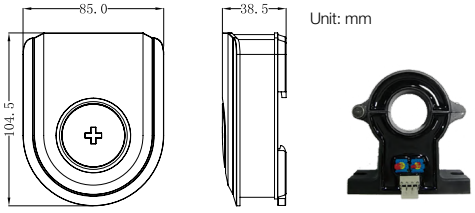
PBMS2000 Battery Management Controller

- Monitor battery string voltage, charge and discharge current
- Calculate battery string SOC
- Auto-balancing
- Auto-sensing for the battery sensor's ID address
- Setpoint alarm (upper limit / lower limit)
- Data collecting
- Data uploading function via built-in RS485 (Modbus-RTU) & Ethernet (Modbus-TCP or SNMP)
- Support 3rd-party system
- Local display via HMI
- Accessories:

  1) Hall sensor and cable: range: -1000A~+1000A (equipped with a 2m cable)
  2) Communication cable : 5m with RJ11 port



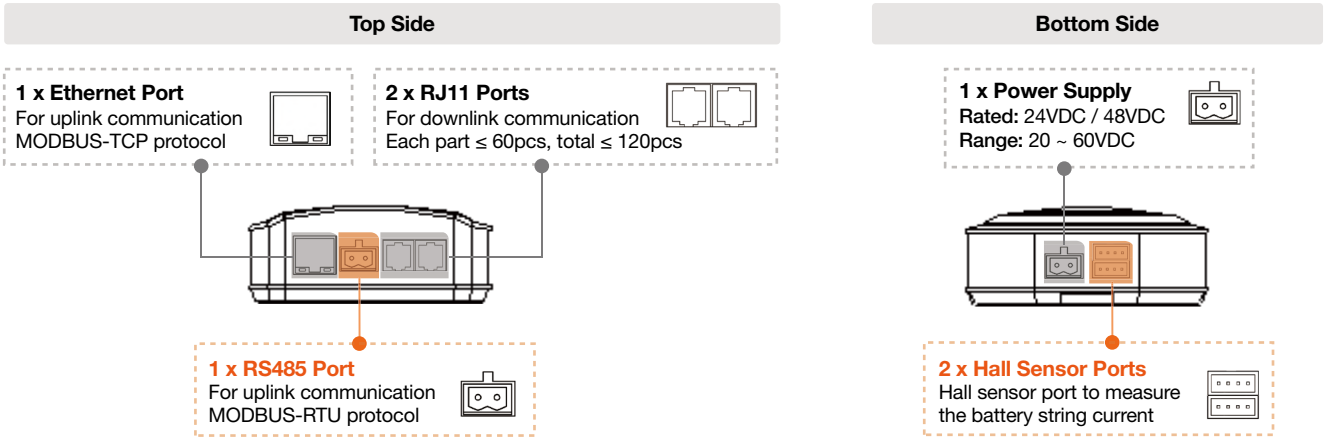
Running Status Alarm Status



Technical Specification

Application		DC power system (telecom site)
Rated Input Voltage		24VDC or 48VDC , range: 20 ~ 60VDC
Measuring	**Voltage**	0 ~ 800VDC (accuracy: ±0.5%)
	Current	DC -1000 ~+1000A (by hall sensor, accuracy: ±2.0%)
Power Comsumption		< 2W
Up-link Communication		1 Etherent port (10/100M), Modbus-TCP / SNMP protocol
		1 RS485 port, Modbus-RTU protocol, baudrate: 9600bps, 19200bps, 38400bps (optional)
Down-link Communication		2 RJ11 ports, each port connect Max. 60pcs batteries, total Max. 120pcs batteries

Terminal Introduction



Battery Cell Sensor

PBAT61-02/PBAT61-06/PBAT61-12 Battery Cell Sensor

- PBAT61-02 for 2V battery, PBAT61-06 for 6V battery, PBAT61-12 for 12V battery
- Monitor individual battery voltage, internal temperature (negative pole), Impedance(ohmic value)
- Auto-balancing



Running Status

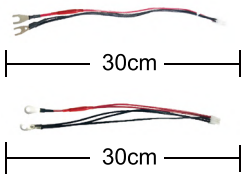


Abnormal Status

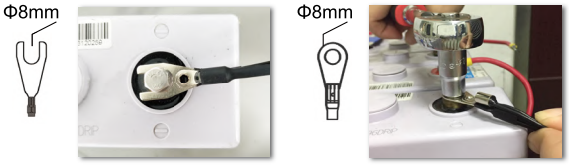
Item	Rated Input Voltage	Measuring Range			
		Voltage	Internal Temperature	Impedance	Power Consumption
PBAT61-02	02V	1.6~2.6V (±0.2%)	-20°C~85°C (±1.0°C)	Range: 0.1~50mΩ Repeatability accuracy: 1.0%±25μΩ Consistency accuracy: 1.5%±25μΩ	Running: <70mW Sleeping: <8mW
PBAT61-06	06V	4.8~10V (±0.2%)			
PBAT61-12	12V	7.5~15.6V (±0.2%)			

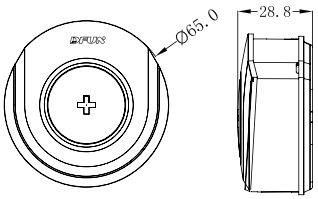
PBMS2000 for Telecom Site

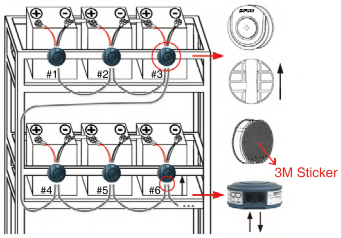
Dimension and Installation

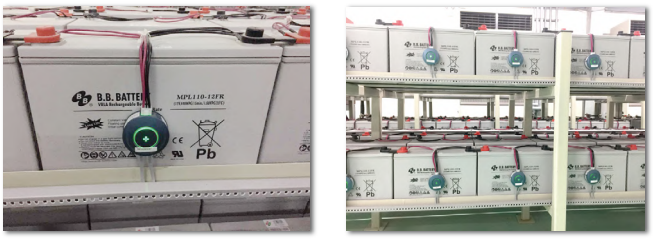


	Item	Description
Battery Measuring Cable	Length	30cm
	Terminal & Size	U type, hole diameter: 8mm
		O type, hole diameter: 8mm









Order Information

System Structure		Model	Description	Remark
Management Layer		PBMS2000	Battery Management Controller 48VDC Power Supply	One PBMS2000 can monitor a maximum of 1 string, totaling 120 batteries, or up to 2 strings, each with a maximum of 60 batteries Hall sensor cable: 2m Communication cable: 5m(with RJ11 port)
Battery Cell Sensor		PBAT 61-02	02V battery cell sensor	One per battery cell
		PBAT 61-12	12V battery cell sensor	One per battery cell
Accessory	Cell Sensor Communication Cable	PBAT-COM-40	Cable terminal: RJ11 port, length: 40cm (standard)	
		PBAT-COM-70	Cable terminal: RJ11 port, length: 70cm	
	Battery Measuring Cable	PBAT-C1-U-M8-30	Cable terminal: U type , hole diameter: 8mm, length: 30cm (standard)	
		PBAT-C1-O-M8-30	Cable terminal: O type , hole diameter: 8mm, length: 30cm	
	Hall Sensor for PBMS2000	CS050EK1T5	Rated input: 50A	Measure Range: -100A~+100A, Φ20mm
		CS100EK2T5	Rated input: 100A	Measure Range: -200A~+200A, Φ40mm
		CS200EK2T5	Rated input: 200A	Measure Range: -400A~+400A, Φ40mm
		CS300EK2T5	Rated input: 300A	Measure Range: -600A~+600A, Φ40mm
		CS400EK2T5	Rated input: 400A	Measure Range: -800A~+800A, Φ40mm
		CS500EK2T5	Rated input: 500A	Measure Range: -1000A~+1000A, Φ40mm
Optional Module	Local Display & Operation	HMI-2000	7-inch touch-screen HMI for local display & operation	One per PBMS2000 Communication with PBMS2000 via RS485

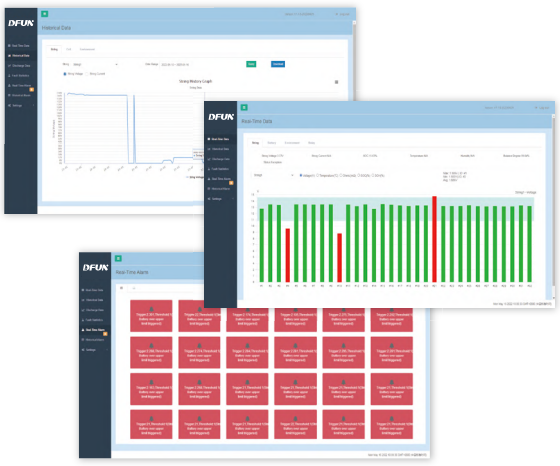
Note: If battery terminal is M5, M6, M10, M12, M16, please contact our sales for customize requirements before placing the order.

PBMS9000
for Data Center



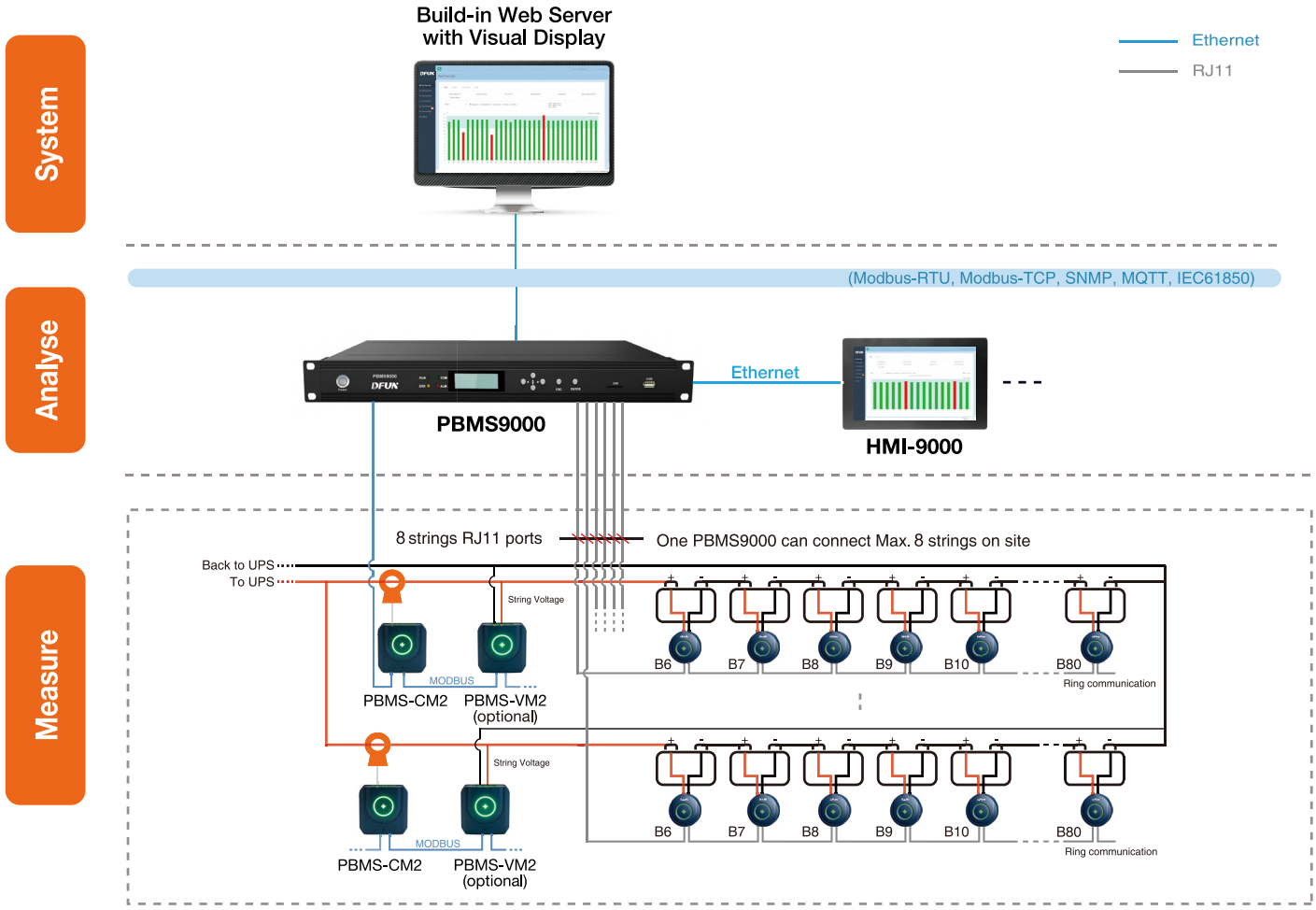
Feature

- Apply to UPS and data center application
- Measure lead-acid or multi-pole battery
- Ring communication, any communication failure will not affect other sensors communication
- Monitor battery voltage, current, impedance, insulation resistance, ripple current & voltage, SOC, SOH, etc.
- Support Modbus, SNMP, MQTT and IEC61850 protocols
- Auto-sensing for the battery sensor's ID address
- Dual-source to avoid power shutdown
- Anti-interference design, support to connect with high-frequency UPS
- Comply with IEEE 1188-2005
- Thermal runaway management(Optional temperature and humidity sensor)
- The product has obtained CE, FCC, Rohs, UL certification



PBMS9000 for Data Center

System Structure



Site 1

Management Layer

PBMS9000 Battery Management Controller

- Standard 1 U design for distribution cabinet
- Dual-source to avoid power shutdown
- Built-in web server with visual display
- Monitor Max. 8 strings battery, in a total of 640 batteries
- Measure battery string current & voltage, ripple voltage & current, charge & discharge current, internal temperature, impedance, insulation resistance, ambient temperature & humidity, SOC and SOH
- Alarm message by SMS or Email
- Support Modbus-TCP, Modbus-RTU, SNMP and IEC61850 protocols
- Support MQTT for JSON format to data upload
- 1 RS485 port, 2 Ethernet ports and 1 4G Antenna port to data upload
- 6 DI ports (digital Input connecting)
- 6 DO ports (sound and light alarm)

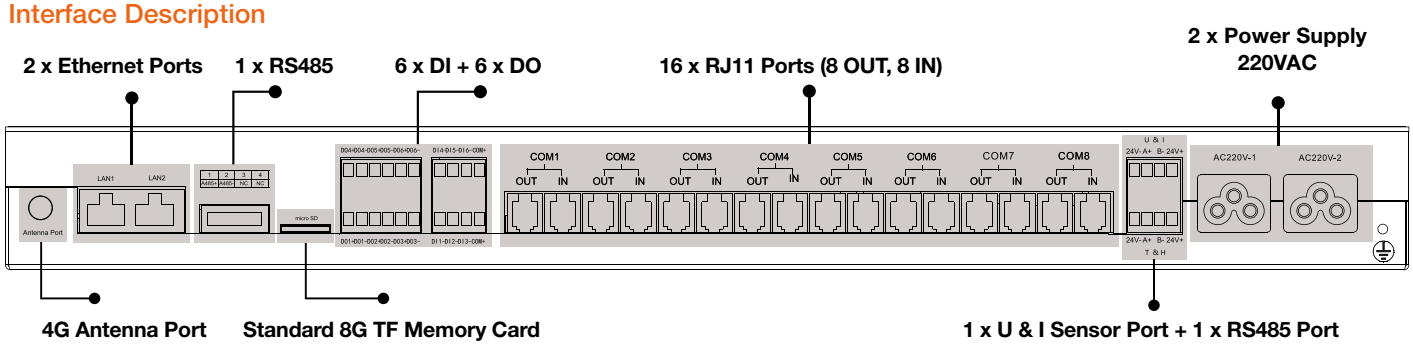


Unit: mm

Pre-install & Pre-wiring for PBMS9000 and HMI-9000 (optional)



Interface Description



Technical Specification

CPU	ARM cortex-A7 528MHz	**Up-link Communication**	2 Ethernet ports (10/100M), MODBUS-TCP, SNMP, IEC61850 1 RS485 port, MODBUS-RTU, baudrate: 9600bps, 19200bps, 38400bps (optional)	
Memory	512MB flash, 4G EMMC + 8G TF memory card	**Down-link Communication**	8 Channels RJ11 ports, each port Max. connect ≤ 80pcs batteries, total Max. 640pcs	
Display	2-inch LCD with backlight	**Measure Range**	Voltage	1~8 strings, Range: 20~1000VDC (±0.1%), Resolution:0.01V
MTBF	≥100,000 hours		Ripple Voltage	1~8 strings, Range: 2~100V (peak), Resolution: 0.01V
Power Supply	V1: 48VDC, Range: 18~72VDC V2: 220VAC, Range: 90~264VAC/127~370VDC V3: 110VDC, Range: 43~160VDC		Current	1~8 strings, Range: -3000~3000A DC (±1%fs, 15°C~35°C), Resolution:0.01A
Dimension	Standard 19-inch 1 U device 483mm × 206mm × 44.5mm (W*D*H) Open hole: 440mm×46mm (L*H)		Ripple Current	1~8 strings, 20% of Hall sensor rated current (peak), Resolution:0.01A
Power Consumption	<15W (only main controller)	**Environment**	Range: -10~+55°C, Resolution:0.1C, Accuracy: ± 0.5 C	
Additional Port	6 x DI dry contact 6 x DO relay output, 250VAC/5A or 30VDC/5A	**Extension Sensor (optional)**	- RS485 for Max. 8 strings ambient temperature & humidity measuring - RS485 for 1~8 strings DC insulation resistance measuring	

String Measuring Sensor

PBMS-CM2 String Current Measuring Sensor & Hall Sensor

- 1 string needs 1 PBMS-CM2, each PBMS-CM2 with 2 hall sensor ports
- Measure battery string charge and discharge current, ripple current
- Hall sensor measuring range: -1000A~+1000A(single-pole) / -2000A~+2000A(dual-pole)
- Supports floating charge hall detection



Hall sensor



Item	Power Supply	Hall Interface	Environment
PBMS-CM2	24VDC (range: 9~32VDC) Power consumption: <0.5W	2 channels, DC ±4V	-10˚C ~ +55˚C , accuracy: ±1˚C

PBMS9000 for Data Center

Battery Cell Sensor

PBAT61-02/PBAT61-06/PBAT61-12 Battery Cell Sensor

- PBAT61-02 for 2V battery, PBAT61-06 for 6V battery, PBAT61-12 for 12V battery
- Monitor individual battery voltage, internal temperature (negative pole), Impedance(ohmic value)
- Auto-balancing



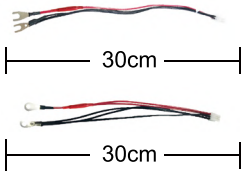
Running Status



Abnormal Status

Item	Rated Input Voltage	Measuring Range			
		Voltage	Internal Temperature	Impedance	Power Consumption
PBAT61-02	02V	1.6~2.6V (±0.2%)	-20°C~85°C (±1.0°C)	Range: 0.1~50mΩ Repeatability accuracy: 1.0%±25μΩ Consistency accuracy: 1.5%±25μΩ	Running: <70mW Sleeping: <8mW
PBAT61-06	06V	4.8~10V (±0.2%)			
PBAT61-12	12V	7.5~15.6V (±0.2%)			

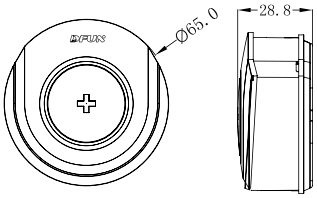
Dimension and Installation

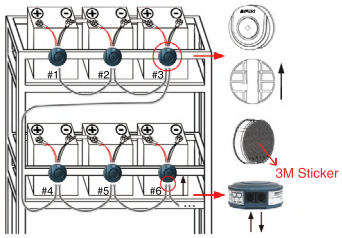


	Item	Description
Battery Measuring Cable	Length	30cm
	Terminal & Size	U type, hole diameter: 8mm
		O type, hole diameter: 8mm

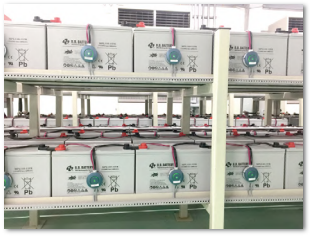












Extension Sensor (Optional)



Ambient Temperature & Humidity Sensor

- Dual RJ45 interface, fast wiring, support cascade use
- Adsorption by magnet
- Temperature: -20°C ~ 60°C (±0.4°C), resolution: 0.1°C. Humidity: 0 ~ 100%RH (±3%RH), resolution: 0.1%RH
- Cable(9000-H-THD-RJ45-500): 5 meter RJ45 convert to 4P terminal wire



Hydrogen (H_2) Sensor

- Class of protection: IP65
- 10~30VDC, power consumption: 0.9W
- Communication: RS485, Modbus
- Measuring range: 0~1000ppm
- Accuracy: ±5%FS
- Resolution: 1ppm(100KΩ~50KΩ)
- Cable: 2 meter Aviation port to 4P terminal wire



PBMS-VM2 String Voltage Measuring Sensor

- 1 PBMS-VM2 can measure 1 string's voltage
- Measure battery string voltage and ripple voltage
- Built-in ground fault detection & testing



DC Leakage Current Sensor

- Monitoring string leakage current to prevent fire caused
- Measurement accuracy: ±1%FS
- Cable: 2 meter hall sensor port to 4P terminal wire

Order Information

System Structure		Model	Description			Remark
Management Layer	BMS Main Controller	PBMS9000-8C- ① - ② - ③ - ④				Monitors Max.8 strings (Max.640 batteries)
		Mandatory	① Power supply voltage	V1	48VDC	
				V2	220VAC	
				V3	110VDC	
			② Power supply channel	1P	Single channel	
				2P	Two channels	
		Optional	③ Communication	L	Support 4G	
			④ Protocol (Choose one option)	CP	Support IEC61850	Monitors Max.108 batteries
				TC	Support IEC104	
String Measuring Sensor		PBMS-CM2	String current measuring sensor			One PBMS-CM2 per string, each is equipped with 2 Hall sensor ports, and can connect a maximum of 2 Hall sensors. If is used for multi-pole batteries, we can configure the PBMS-CM2 according to the number of poles Hall sensor cable: 2m Communication cable: 5m(with RJ45 port)
		PBMS-VM2	String voltage measuring sensor			One per string (optional)
Battery Cell Sensor		PBAT 61-02	02V battery cell sensor			One per battery cell
		PBAT 61-06	06V battery cell sensor			One per battery cell
		PBAT 61-12	12V battery cell sensor			One per battery cell
Accessory	Cell Sensor Communication Cable	PBAT-COM-40	Cable terminal: RJ11 port, length: 40cm (standard)			
		PBAT-COM-70	Cable terminal: RJ11 port, length: 70cm			
	Battery Measuring Cable	PBAT-C1-U-M8-30	Cable terminal: U type  , hole diameter: 8mm, length: 30cm (standard)			
		PBAT-C1-O-M8-30	Cable terminal: O type , hole diameter: 8mm, length: 30cm			
	Hall Sensor for PBMS-CM2	DXEEK03-50A-V4	Rated input: 50A			Measure Range: -50A~+50A, Φ20mm
		DXEEK03-100A-V4	Rated input: 100A			Measure Range: -100A~+100A, Φ20mm
		DXEEK05-25A-V4	Rated input: 25A			Measure Range: -25A~+25A, Φ40mm
		DXEEK05-200A-V4	Rated input: 200A			Measure Range: -200A~+200A, Φ40mm
		DXEEK05-300A-V4	Rated input: 300A			Measure Range: -300A~+300A, Φ40mm
		DXEEK05-400A-V4	Rated input: 400A			Measure Range: -400A~+400A, Φ40mm
		DXEEK05-500A-V4	Rated input: 500A			Measure Range: -500A~+500A, Φ40mm
		DXEEK05-600A-V4	Rated input: 600A			Measure Range: -600A~+600A, Φ40mm
		DXEEK05-800A-V4	Rated input: 800A			Measure Range: -800A~+800A, Φ40mm
		DXEEK05-1000A-V4	Rated input: 1000A			Measure Range: -1000A~+1000A, Φ40mm
Optional Module	HMI for Local Display & Operation	HMI-9000	10-inch touch-screen HMI for local display & operation			One per six (max) PBMS9000 Communication with PBMS9000 via Ethernet
		HDR-30-12	12VDC power coverter (220VAC to 12VDC)			If selected HMI-PBMS9000, then must select power supply converter
	DC Leakage Current Sensor	STL8S-MT2-300mA-DF	Rated Input: ±300mA			Measure Range:-360mA~+360mA, Φ72mm
	Ambient Temperature & Humidity	H-THDRJ45	Temperature: -20°C ~ 60°C (±0.4°C) Humidity: 0 ~ 100%RH (±3%RH)			One PBMS9000 can conncet Max. 6pcs
	Hydrogen (H_2) Sensor	PBMS-H_2	Measuring Range: 0~1000ppm			One per one PBMS9000
	Cabinet for Prewiring In Factory	PBMS-BOX-260	Dimension: 535*260*335mm (W*H*D)			Pre-install & pre-wiring for PBMS9000 and HMI-9000
		PBMS-BOX-730	Dimension: 535*730*335mm (W*H*D)			Pre-install & pre-wiring for Max.4 PBMS9000 and one HMI-9000
	Ethernet Switch	IES318	1. Power supply 24VDC 12~48VDC 2. 8 x 10/100M Ethernet ports			If selected PBAT-BOX-730, then must select IES318

Note:

① If battery terminal is M5, M6, M10, M12, M16, please contact our sales for customize requirements before placing the order.

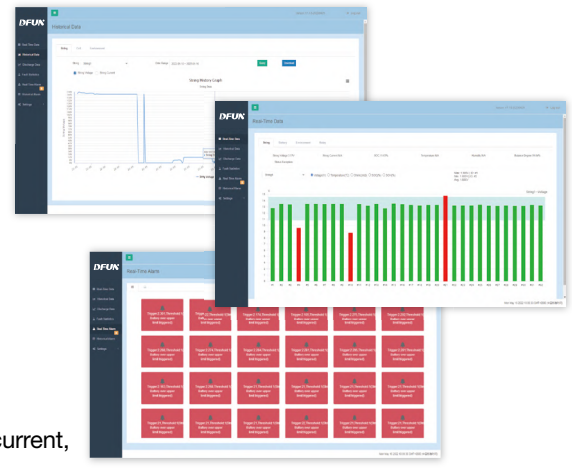
② If the Hall sensor measuring range exceeds -1000~+1000A, please contact our sales for customization before ordering.

PBMS2000Pro
for Power and Rail Industries



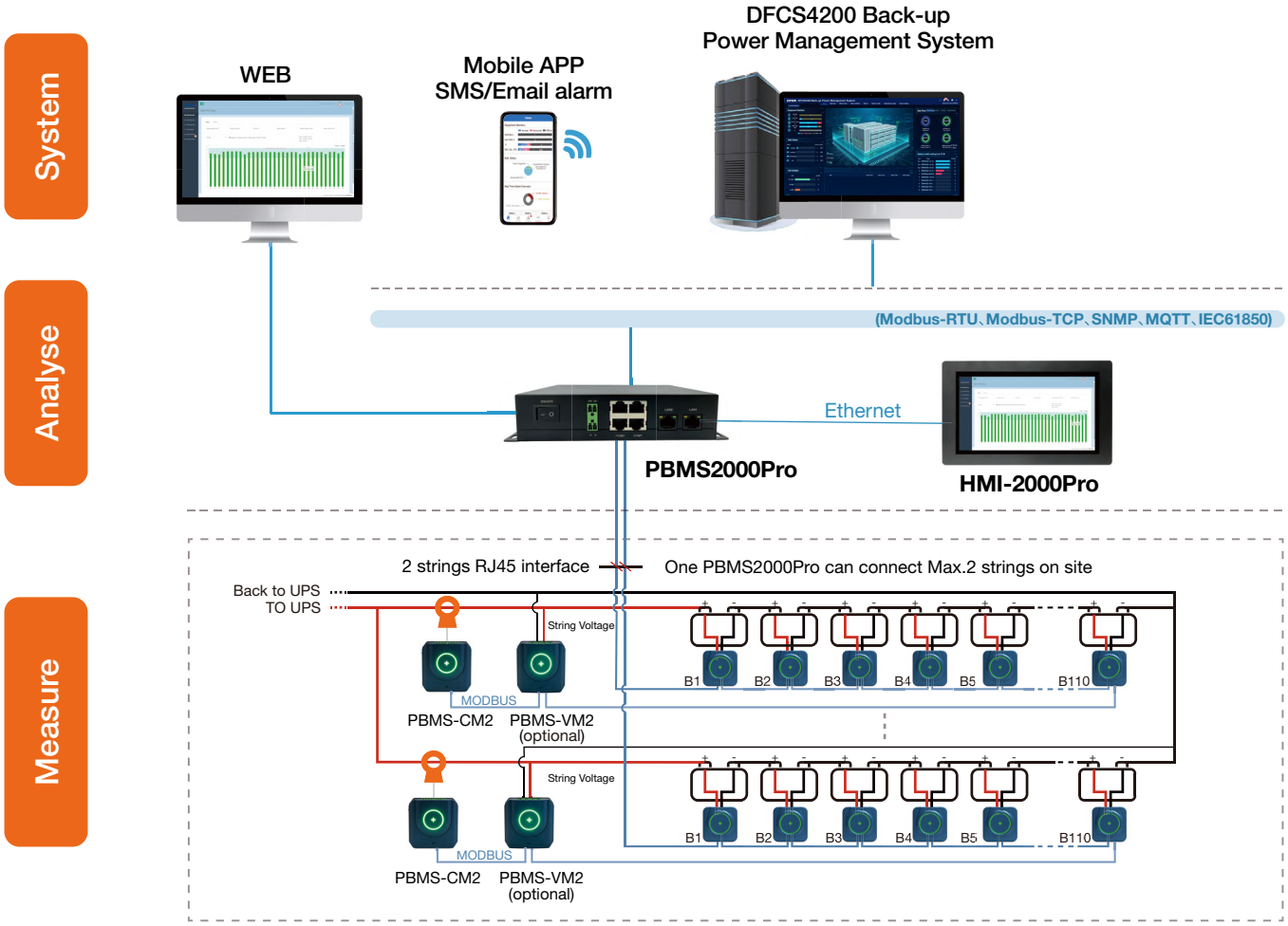
Feature

- Suitable for power systems, rail transportation, semiconductor industry backup power scenarios
- 24/7 online monitoring & remote alarm notifications
- Monitor Max. 2 strings and 220pcs batteries in total
- Measure Ni-Cd, Lead-Acid, and muti-pole battery
- Powered by communication bus, no consuming battery power
- Anti-interference design, support to connect with high-frequency UPS
- Anti-reverse protection, no damaging the cell sensor and battery
- Auto-sensing for the battery sensor's ID address
- Support Modbus, SNMP, MQTT and IEC61850 protocols(optional)
- Support 4G
- Support monitoring battery leakage and liquid level (optional)
- Support extension sensor for measuring ambient temperature & humidity, leakage current, insulation resistance (optional)
- Comply with IEEE 1188-2005
- Thermal runaway management(Optional temperature and humidity sensor)



PBMS2000Pro for Power and Rail Industries

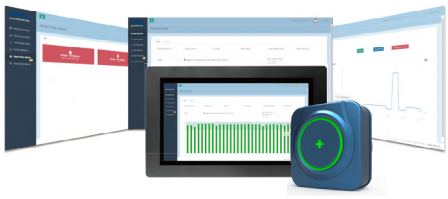
System Structure



Management Layer

PBMS2000Pro Battery Management Controller

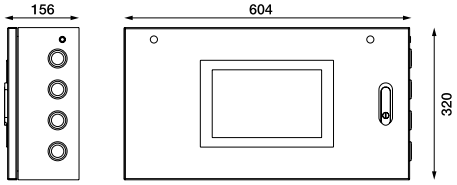
- 7*24 real-time monitoring and remote alerting
- Built-in web server with visual display
- Monitor Max. 2 strings battery, in a total of 220 batteries
- Measure battery string current & voltage, ripple voltage & current, charge & discharge current, internal temperature, impedance, insulation resistance, leakage current, ambient temperature & humidity, SOC and SOH
- Alarm notifications via SMS or Email
- Support Modbus-TCP, Modbus-RTU, SNMP
- 2 RS485 port, 2 Ethernet ports
- 2 DI ports (for digital Input connecting)
- 2 DO ports (for sound and light alarms)
- Feature RIG-B code (DC) time calibration functionality with a synchronization error of less than 1 ms
- 4G antenna port to data upload(optional)
- IEC61850 protocols(optional)





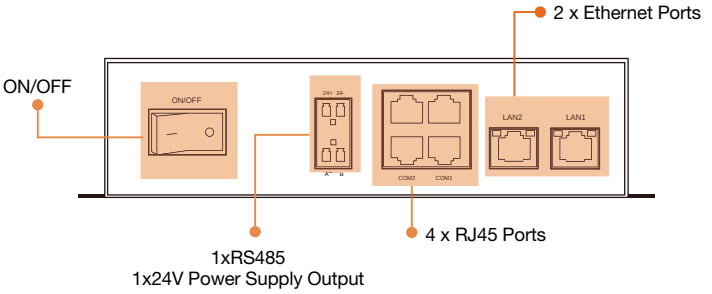


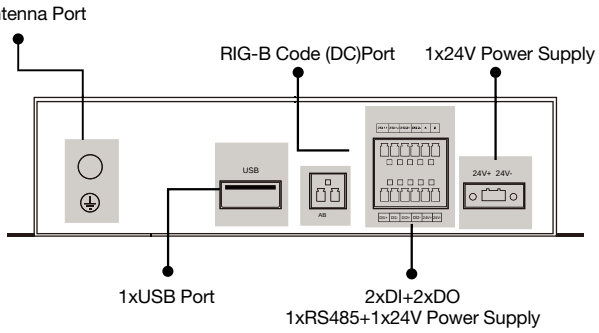
Unit: mm



PBMS2000Pro for Power and Rail Industries

Interface Description





Technical Specification

CPU	Dual-core Cortex-A7, 1.2GHz	**Memory**	128MB DDR3
Memorizer	4GB EMMC	**Additional**	2xEthernet, 2xRS485, 2xDI, 2xDO
Up-link Communication	2 Ethernet Ports,MODBUS-TCP, SNMP,1 RS485	**Down-link Communication**	2 channels RJ45 ports
Dimension	195×120×40mm (L*W*H)	**Measure**	Max. 2 strings battery, Max. 220 batteries

String Measuring Sensor

PBMS-CM2 String Current Measuring Sensor & Hall Sensor

- 1 string needs 1 PBMS-CM2, each PBMS-CM2 with 2 hall sensor ports
- Measure battery string charge and discharge current, ripple current
- Hall sensor measuring range: -1000A~+1000A(single-pole) / -2000A~+2000A(dual-pole)
- Supports floating charge hall detection



Hall sensor



Item	Power Supply	Hall Interface	Environment
PBMS-CM2	24VDC (range: 9~32VDC) Power consumption: <0.5W	2 channels, DC ±4V	-10˚C ~ +55˚C , accuracy: ±1˚C

Battery Cell Sensor

PBAT71-02 / PBAT71-12 Battery Cell Sensor

- PBAT71-02 for 2V Lead-Acid battery or 1.2V Ni-Cd battery
  PBAT71-12 for 12V Lead-Acid battery
- Monitor individual battery voltage, internal temperature (negative pole), impedance(ohmic value)
- Calculate individual battery state of charge (SOC), state of health (SOH)
- Support battery leakage and liquid level monitoring (optional)
- Powered by communication bus, no consuming battery power
- Auto-balancing



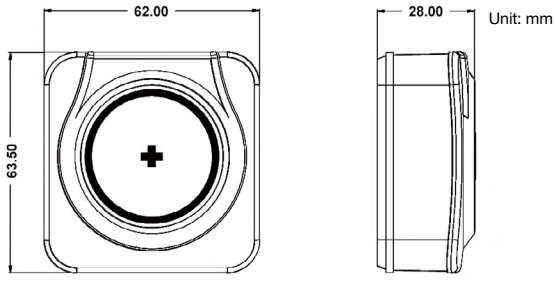
Running Status

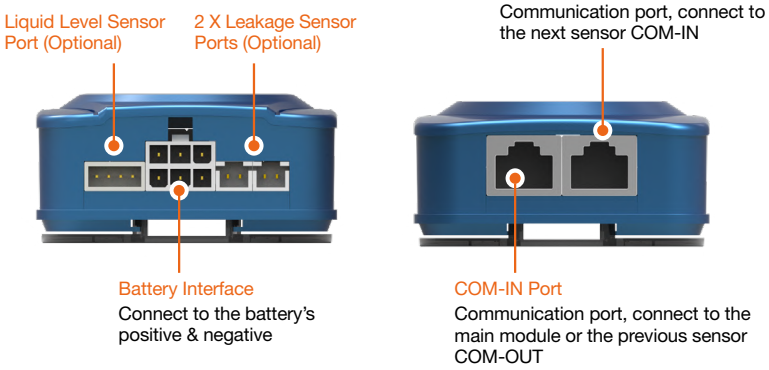


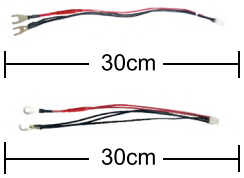
Abnormal Status

Item	Power Supply	Rated Input Voltage	Measuring Range		
			Voltage	Internal Temperature	Impedance
PBAT71-02	24VDC Power Consumption: < 0.25W	02V / 1.2V	0.5~3VDC (±0.2%)	-20°C~85°C (±0.5°C)	Range: 0.1~50mΩ Repeatability accuracy: ±1.0%±25μΩ Consistency accuracy: ±1.5%±25μΩ
PBAT71-12		12V	5~18VDC (±0.2%)		

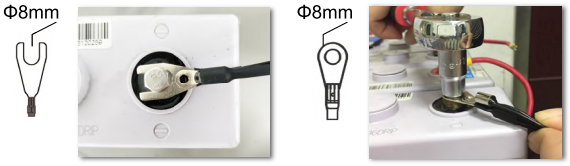
Dimension and Installation







	Item	Description
Battery Measuring Cable	Length	30cm
	Terminal & Size	U type, hole diameter: 8mm
		O type, hole diameter: 8mm



Extension Sensor (Optional)



Ambient Temperature & Humidity Sensor

- Dual RJ45 interface, fast wiring, support cascade use
- Adsorption by magnet
- Temperature: -20°C ~ 60°C (±0.4°C), resolution: 0.1°C. Humidity: 0 ~ 100%RH (±3%RH), resolution: 0.1%RH
- Cable(9000-H-THD-RJ45-500): 5 meter RJ45 convert to 4P terminal wire



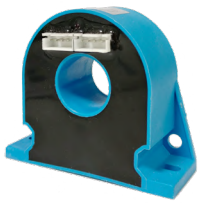
PBMS-VM2 String Voltage Measuring Sensor

- 1 PBMS-VM2 can measure 1 string's voltage
- Measure battery string voltage and ripple voltage
- Built-in ground fault detection & testing



Hydrogen (H_2) Sensor

- Class of protection: IP65
- 10~30VDC, power consumption: 0.9W
- Communication: RS485, Modbus
- Measuring range: 0~1000ppm
- Accuracy: ±5%FS
- Resolution: 1ppm(100KΩ~50KΩ)
- Cable: 2 meter Aviation port to 4P terminal wire



DC Leakage Current Sensor

- Monitoring string leakage current to prevent fire caused
- Measurement accuracy: ±1%FS
- Cable: 2 meter hall sensor port to 4P terminal wire

Order Information

System Structure		Model	Description	Remark
Management Layer	BMS Main Controller	PBMS2000Pro	Battery Management Controller	One per UPS One PBMS2000Pro can monitor Max.2 battery strings (Max.220 batteries)
		PBMS2000Pro-L	Battery Management Controller with 4G antenna port	
	Protocol(optional)	PBMS2000Pro-CP	DFUN standard IEC61850 model available. For customized needs, please contact our sales team.	
String Measuring Sensor		PBMS-CM2	String current measuring sensor	Each has 2 hall sensor ports,one or two PBMS-CM2 per string Hall sensor cable: 2m Communication cable: 5m with RJ45 port
		PBMS-VM2	String voltage measuring sensor	One per string (optional)
Battery Cell Sensor		PBAT 71-02	02V / 1.2V battery cell sensor	One per battery cell
		PBAT 71-12	12V battery cell sensor	One per battery cell
Accessory	Cell Sensor Communication Cable	PBAT-COM-RJ45-20	Cable terminal: RJ45 port, length: 20cm	
		PBAT-COM-RJ45-50	Cable terminal: RJ45 port, length: 50cm (standard)	
	Battery Measuring Cable	PBAT-C1-U-M8-30	Cable terminal: U type  , hole diameter: 8mm, length: 30cm (standard)	
		PBAT-C1-O-M8-30	Cable terminal: O type  , hole diameter: 8mm, length: 30cm	
	Hall Sensor for PBMS-CM2	DXEEK03-50A-V4	Rated input: 50A	Measure Range: -50A~+50A, Φ20mm
		DXEEK03-100A-V4	Rated input: 100A	Measure Range: -100A~+100A, Φ20mm
		DXEEK05-25A-V4	Rated input: 25A	Measure Range: -25A~+25A, Φ40mm
		DXEEK05-200A-V4	Rated input: 200A	Measure Range: -200A~+200A, Φ40mm
		DXEEK05-300A-V4	Rated input:300A	Measure Range: -300A~+300A, Φ40mm
		DXEEK05-400A-V4	Rated input: 400A	Measure Range: -400A~+400A, Φ40mm
		DXEEK05-500A-V4	Rated input: 500A	Measure Range: -500A~+500A, Φ40mm
		DXEEK05-600A-V4	Rated input: 600A	Measure Range: -600A~+600A, Φ40mm
		DXEEK05-800A-V4	Rated input: 800A	Measure Range: -800A~+800A, Φ40mm
		DXEEK05-1000A-V4	Rated input: 1000A	Measure Range: -1000A~+1000A, Φ40mm
	Power Supply	HDR-150-24	220VDC/220VAC to 24VDC power coverter	One per PBMS2000Pro,Max. 220 batteries
		SD-100D-24	110VDC to 24VDC power coverter	One per PBMS2000Pro,Max. 110 batteries
		RSD-100C-24	48VDC to 24VDC power coverter	One per PBMS2000Pro,Max. 110 batteries
Optional Module	HMI for Local Display & Operation	HMI-2000Pro	10-inch touch-screen HMI for local display	One per 6 PBMS2000Pro maxium Communication with PBMS2000Pro via Ethernet
	DC Leakage Current Sensor	STL8S-MT2-300mA-DF	Rated input: ±300mA	Measure Range:-360mA~+360mA, Φ72mm
	Ambient Temperature & Humidity	H-THDRJ45	Temperature: -20°C~60°C (±0.4°C) Humidity: 0~100%RH (±3%RH)	1 PBMS2000Pro can conncet Max. 2pcs
	Hydrogen (H_2) Sensor	PBMS-H_2	Measuring range: 0~1000ppm	One per one PBMS2000Pro
	Liquid Level Sensor	PBMS-ELS	Monitor liquid level, for transparent battery	One per battery, length: 30cm
		PBMS-Y29	Monitor liquid level, for non-transparent battery	One per battery, length: 50cm
	Leakage Sensor	PBMS-LS-O35	Monitor leakage with two ports (connect to the battery +/- port)	One/Two per battery, length: 30cm Hole diameter: 35mm
		PBMS-LS-O41	Monitor leakage with two ports (connect to the battery +/- port)	One/Two per battery, length: 30cm Hole diameter: 41mm

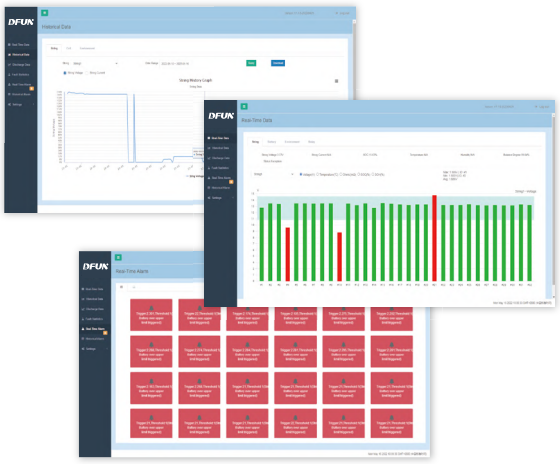
Note: If battery terminal is M5, M6, M10, M12, M16, please contact our sales for customize requirements before placing the order.

PBMS9000Pro
for Substation



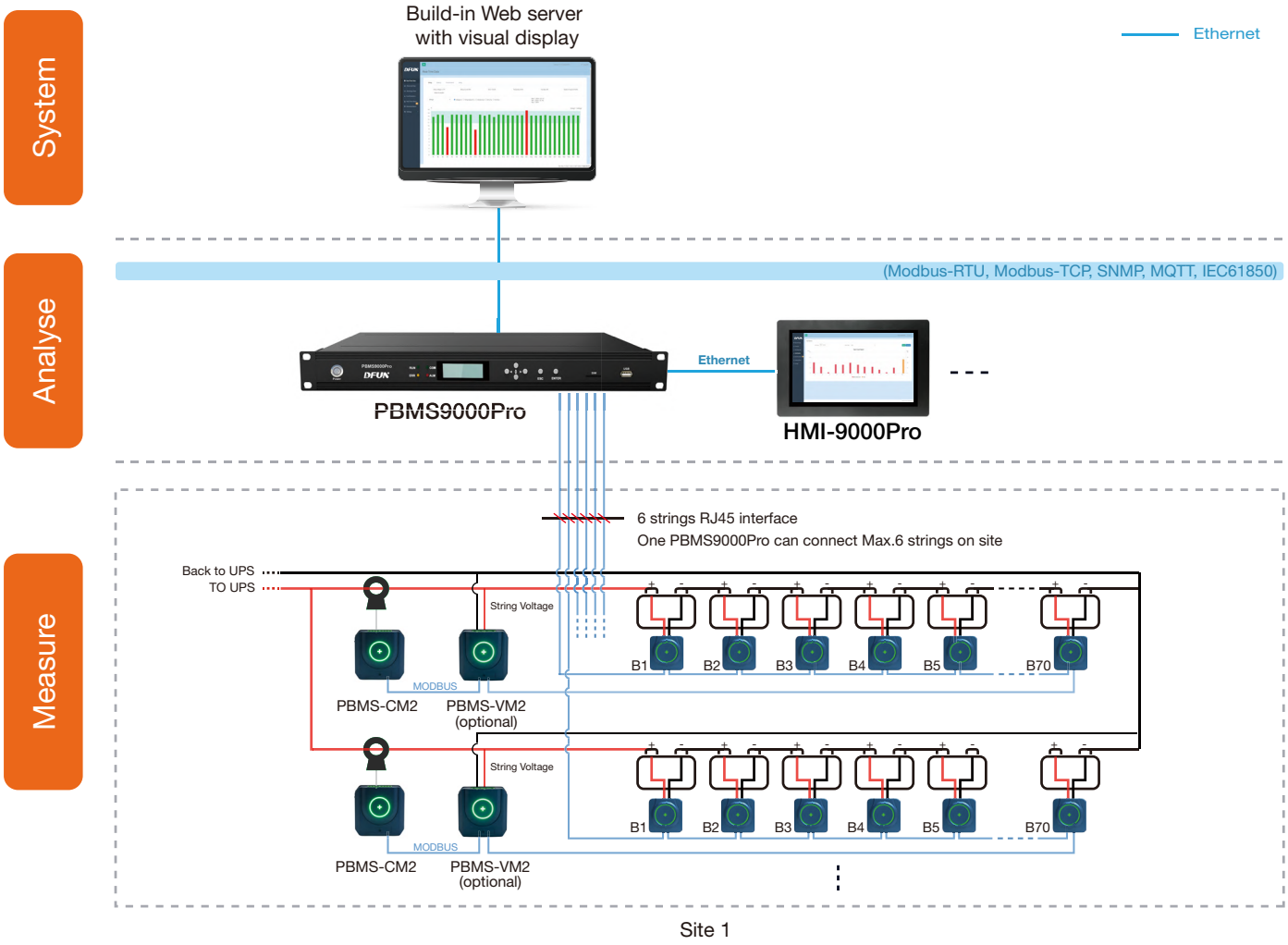
Feature

- Apply to DC power systems like substation, power plants, etc
- 24/7 online monitoring & remote alarm notifications
- Monitor Max. 6 strings and 420pcs batteries in total
- Measure Ni-Cd, Lead-Acid, and muti-pole battery
- Powered by communication bus, no consuming battery power
- Anti-interference design, support to connect with high-frequency UPS
- Anti-reverse protection, no damaging the cell sensor and battery
- Auto-sensing for the battery sensor's ID address
- Support Modbus, SNMP, MQTT and IEC61850 protocols
- Support monitoring battery leakage and liquid level (optional)
- Support extension sensor for measuring ambient temperature & humidity, leakage current, insulation resistance (optional)
- Comply with IEEE 1188-2005
- Thermal runaway management(Optional temperature and humidity sensor)



PBMS9000Pro for Substation

System Structure

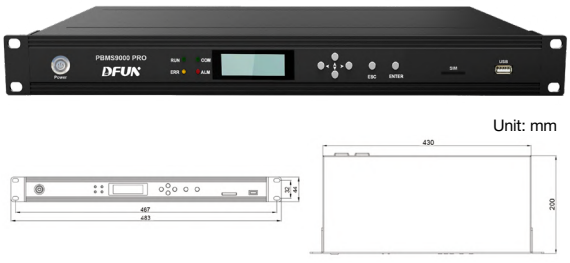


Site 1

Management Layer

PBMS9000Pro Battery Management Controller

- Standard 1 U design for distribution cabinet
- Dual-source to avoid power shutdown
- Built-in web server with visual display
- Monitor Max. 6 battery strings, in a total of 420 batteries
- Measure battery string current & voltage, ripple voltage & current, charge & discharge current, internal temperature, impedance, insulation resistance, ambient temperature & humidity, SOC and SOH
- Alarm notifications via SMS or Email
- Support Modbus-TCP, Modbus-RTU, SNMP and IEC61850 protocols
- Support MQTT for JSON format to data upload
- 1 RS485 port, 2 Ethernet ports and 1 4G antenna port to data upload
- 6 DI ports (for digital Input connecting)
- 6 DO ports (for sound and light alarms)

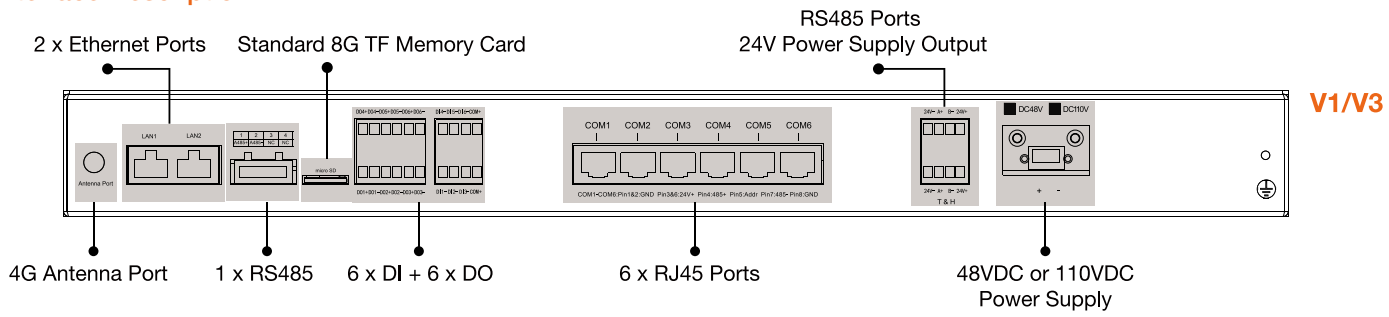


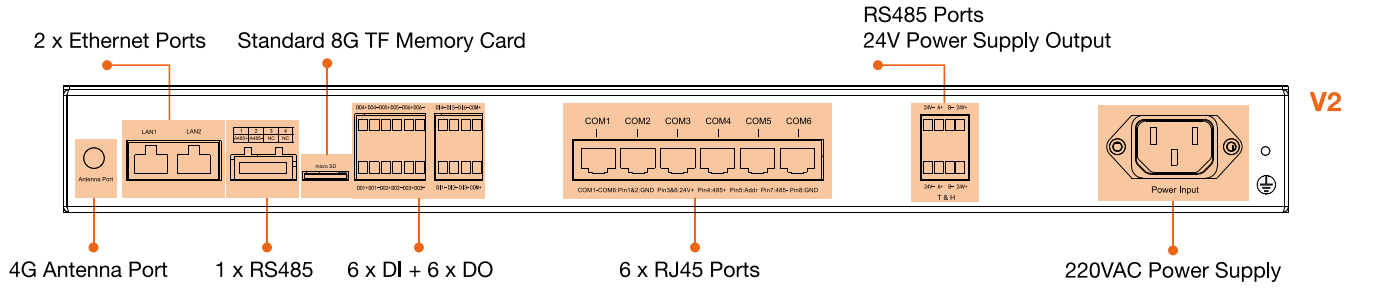
Unit: mm

Pre-install & Pre-wiring for PBMS9000Pro and HMI-9000 (optional)



Interface Description





Technical Specification

CPU	ARM cortex-A7 528MHz	**Up-link Communication**	2 Ethernet ports (10/100M), MODBUS-TCP, SNMP, IEC61850 1 RS485 port, MODBUS-RTU, baudrate: 9600bps, 19200bps, 38400bps (optional)	
Memory	512MB flash, 4G EMMC + 8G TF memory card			
Display	2-inch LCD with backlight	**Down-link Communication**	6 channels RJ45 ports,each port Max. connect 70pcs batteries	
MTBF	≥100,000 hours	**Measure Range**	Voltage	1~6 strings, Range: 20~1000V DC (±0.5%), Resolution:0.01V
Power Supply	V1: 48VDC, Range: 18~72VDC V2: 220VAC, Range: 90~264VAC/127~370VDC V3: 110VDC, Range: 43~160VDC		Current	1~6 strings, Max. Range: -1000~+1000A DC (±1.0%fs,15°C~30°C), Resolution: 0.01A
Dimension	Standard 19-inch 1 U device 483mm × 206mm × 44.5mm (W*D*H) Open hole: 440mm×46mm (L*H)	**Operation Environment**	Operating Temperature: -10~65°C (±1°C) Storage Temperature: -40~70°C Humidity: 10~95% RH, non-condensing	
Power Consumption	<15W (Only main controller)			
Additional Port	6 x DI dry contact 6 x DO relay output, 250VAC/5A or 30VDC/5A	**Extension Sensor (optional)**	-RS485 for Max. 6 strings ambient temperature & humidity measuring -RS485 for 1~6 strings DC leakage current measuring -RS485 for 1~6 strings DC insulation resistance measuring	

Battery Cell Sensor

PBAT71-02 / PBAT71-12 Battery Cell Sensor

- PBAT71-02 for 2V Lead-Acid battery or 1.2V Ni-Cd battery
  PBAT71-12 for 12V Lead-Acid battery
- Monitor individual battery voltage, internal temperature (negative pole), impedance(ohmic value)
- Calculate individual battery state of charge (SOC), state of health (SOH)
- Support battery leakage and liquid level monitoring (optional)
- Powered by communication bus, no consuming battery power
- Auto-balancing

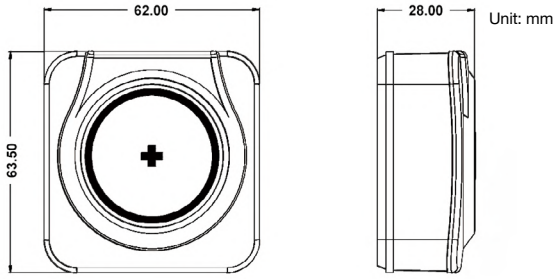

Running Status

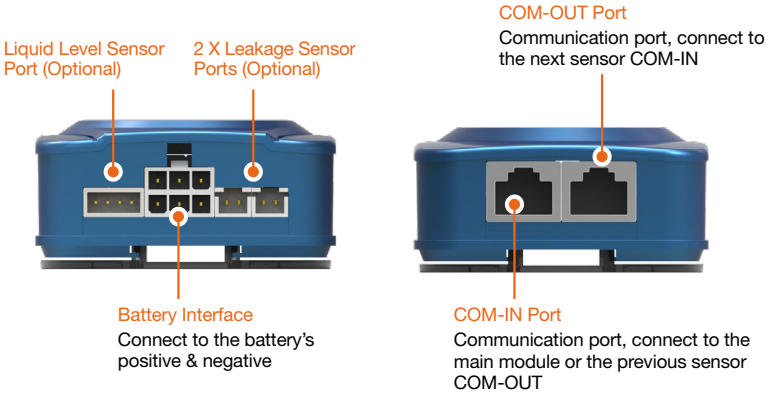

Abnormal Status

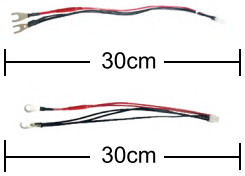
Item	Power Supply	Rated Input Voltage	Measuring Range		
			Voltage	Internal Temperature	Impedance
PBAT71-02	24VDC Power Consumption: < 0.25W	02V / 1.2V	0.5~3VDC (±0.2%)	-20°C~85°C (±0.5°C)	Range: 0.1~50mΩ Repeatability accuracy: ±1.0%±25μΩ Consistency accuracy: ±1.5%±25μΩ
PBAT71-12		12V	5~18VDC (±0.2%)		

PBMS9000Pro for Substation

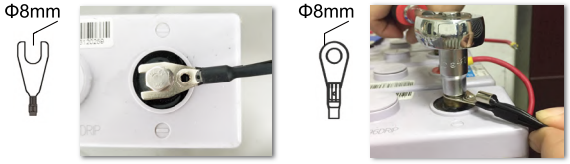
Dimension and Installation







	Item	Description
Battery Measuring Cable	Length	30cm
	Terminal & Size	U type, hole diameter: 8mm
		O type, hole diameter: 8mm



String Measuring Sensor

PBMS-CM2 String Current Measuring Sensor & Hall Sensor

- 1 string needs 1 PBMS-CM2, each PBMS-CM2 with 2 hall sensor ports
- Measure battery string charge and discharge current, ripple current
- Hall sensor measuring range: -1000A~+1000A(single-pole) / -2000A~+2000A(dual-pole)
- Supports floating charge hall detection



Hall sensor



Item	Power Supply	Hall Interface	Environment
PBMS-CM2	24VDC (range: 9~32VDC) Power consumption: <0.5W	2 channels, DC ±4V	-10˚C ~ +55˚C , accuracy: ±1˚C

Extension Sensor (Optional)



Ambient Temperature & Humidity Sensor

- Dual RJ45 interface, fast wiring, support cascade use
- Adsorption by magnet
- Temperature: -20°C ~ 60°C (±0.4°C), resolution: 0.1°C. Humidity: 0 ~ 100%RH (±3%RH), resolution: 0.1%RH
- Cable(9000-H-THD-RJ45-500): 5 meter RJ45 convert to 4P terminal wire



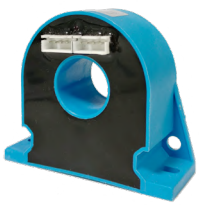
PBMS-VM2 String Voltage Measuring Sensor

- 1 PBMS-VM2 can measure 1 string's voltage
- Measure battery string voltage and ripple voltage
- Built-in ground fault detection & testing



Hydrogen (H_2) Sensor

- Class of protection: IP65
- 10~30VDC, power consumption: 0.9W
- Communication: RS485, Modbus
- Measuring range: 0~1000ppm
- Accuracy: ±5%FS
- Resolution: 1ppm(100KΩ~50KΩ)
- Cable: 2 meter Aviation port to 4P terminal wire



DC Leakage Current Sensor

- Monitoring string leakage current to prevent fire caused
- Measurement accuracy: ±1%FS
- Cable: 2 meter hall sensor port to 4P terminal wire

Order Information

System Structure		Model	Description	Remark
Management Layer	BMS Main Controller	PBMS9000Pro-V1	Battery Management Controller, 48VDC power supply	One per UPS One PBMS9000Pro can monitor Max.6 battery strings (Max.100 batteries)
		PBMS9000Pro-V1-L	Battery Management Controller, 48VDC power supply with 4G	
		PBMS9000Pro-V2	Battery Management Controller, 220VAC power supply	One per UPS One PBMS9000Pro can monitor Max.6 battery strings (Max.420 batteries)
		PBMS9000Pro-V2-L	Battery Management Controller, 220VAC power supply with 4G	
		PBMS9000Pro-V3	Battery Management Controller, 110VDC power supply	One per UPS One PBMS9000Pro can monitor Max.6 battery strings (Max.216 batteries)
		PBMS9000Pro-V3-L	Battery Management Controller, 110VDC power supply with 4G	
	Protocol(optional)	PBMS9000Pro-CP	DFUN standard IEC61850 model available. For customized needs, please contact our sales team.	
String Measuring Sensor		PBMS-CM2	String current measuring sensor	Each has 2 hall sensor ports, one or two PBMS-CM2 per string Hall sensor cable: 2m cable Communication cable: 5m with RJ45 port
		PBMS-VM2	String voltage measuring sensor	One per each string (optional)
Battery Cell Sensor		PBAT 71-02	02V / 1.2V battery cell sensor	One per battery cell
		PBAT 71-12	12V battery cell sensor	One per battery cell
Accessory	Cell Sensor Communication Cable	PBAT-COM-RJ45-20	Cable terminal: RJ45 port, length: 20cm	
		PBAT-COM-RJ45-50	Cable terminal: RJ45 port, length: 50cm (standard)	
	Battery Measuring Cable	PBAT-C1-U-M8-30	Cable terminal: U type  , hole diameter: 8mm, length: 30cm (standard)	
		PBAT-C1-O-M8-30	Cable terminal: O type  , hole diameter: 8mm, length: 30cm	
	Hall Sensor for PBMS-CM2	DXEEK03-50A-V4	Rated input: 50A	Measure Range: -50A~+50A, Φ20mm
		DXEEK03-100A-V4	Rated input: 100A	Measure Range: -100A~+100A, Φ20mm
		DXEEK05-25A-V4	Rated input: 25A	Measure Range: -25A~+25A, Φ40mm
		DXEEK05-200A-V4	Rated input: 200A	Measure Range: -200A~+200A, Φ40mm
		DXEEK05-300A-V4	Rated input:300A	Measure Range: -300A~+300A, Φ40mm
		DXEEK05-400A-V4	Rated input: 400A	Measure Range: -400A~+400A, Φ40mm
		DXEEK05-500A-V4	Rated input: 500A	Measure Range: -500A~+500A, Φ40mm
		DXEEK05-600A-V4	Rated input: 600A	Measure Range: -600A~+600A, Φ40mm
		DXEEK05-800A-V4	Rated input: 800A	Measure Range: -800A~+800A, Φ40mm
		DXEEK05-1000A-V4	Rated input: 1000A	Measure Range: -1000A~+1000A, Φ40mm
Optional Module	HMI for Local Display & Operation	HMI-9000Pro	10-inch touch-screen HMI for local display & operation	Power Supply from Auxiliary power module
	Auxiliary Power Module (choose one out of three)	HDR-30-12	220V AC to 12V DC power supply	Provide power supply for the HMI-9000Pro
		DDR-15L-12	48V DC to 12V DC power supply	
		RSD-30H-12	110V DC to 12V DC power supply	
	DC Leakage Current Sensor	STL8S-MT2-300mA-DF	Rated input: ±300mA	Measure range: -360mA~+360mA, Φ72mm
	Ambient Temperature & Humidity	H-THDRJ45	Temperature: -20~60°C (±0.4°C) Humidity: 0 ~ 100%RH (±3%RH)	1 PBMS9000Pro can conncet Max. 6pcs
	Hydrogen (H_2) Sensor	PBMS-H_2	Measuring range: 0 ~ 1000ppm	One per one PBMS9000Pro
	Liquid Level Sensor	PBMS-ELS	Monitor liquid level, for transparent battery	One per battery, length: 30cm
		PBMS-Y29	Monitor liquid level, for non-transparent battery	One per battery, length: 50cm
	Leakage Sensor	PBMS-LS-O35	Monitor leakage with two ports (connect to the battery +/- port)	One/Two per each battery, length: 30cm Hole diameter: 35mm
		PBMS-LS-O41	Monitor leakage with two ports (connect to the battery +/- port)	One/Two per each battery, length: 30cm Hole diameter: 41mm
	Cabinet for Prewiring In Factory	PBMS-BOX-260	Dimension: 535*260*335mm (W*H*D)	Pre-install & pre-wiring for PBMS9000Pro and HMI-9000Pro
		PBMS-BOX-730	Dimension: 535*730*335mm (W*H*D)	Pre-install & pre-wiring for Max.4 PBMS9000Pro and one HMI-9000Pro
	Ethernet Switch	IES318	1. Power supply 24VDC 12~48VDC 2. 8 x 10/100M Ethernet ports	If selected PBAT-BOX-730, then must select IES318

Note: If battery terminal is M5, M6, M10, M12, M16, please contact our sales for customize requirements before placing the order.

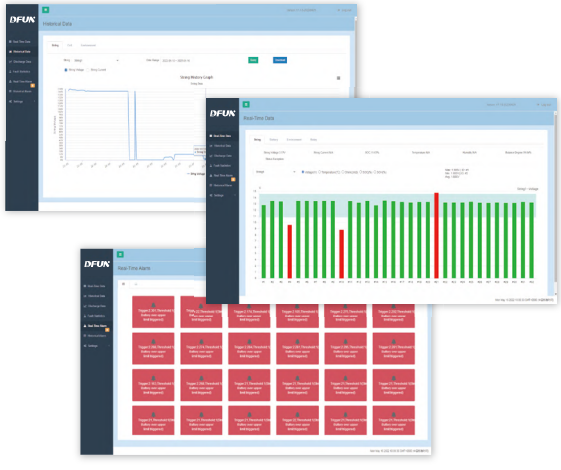
PBMS9000Pro
for Chemical Plant



- Monitor Ni-Cd & Lead-Acid Battery
- IP65 Protection
- UL94-V0 Fire Rating

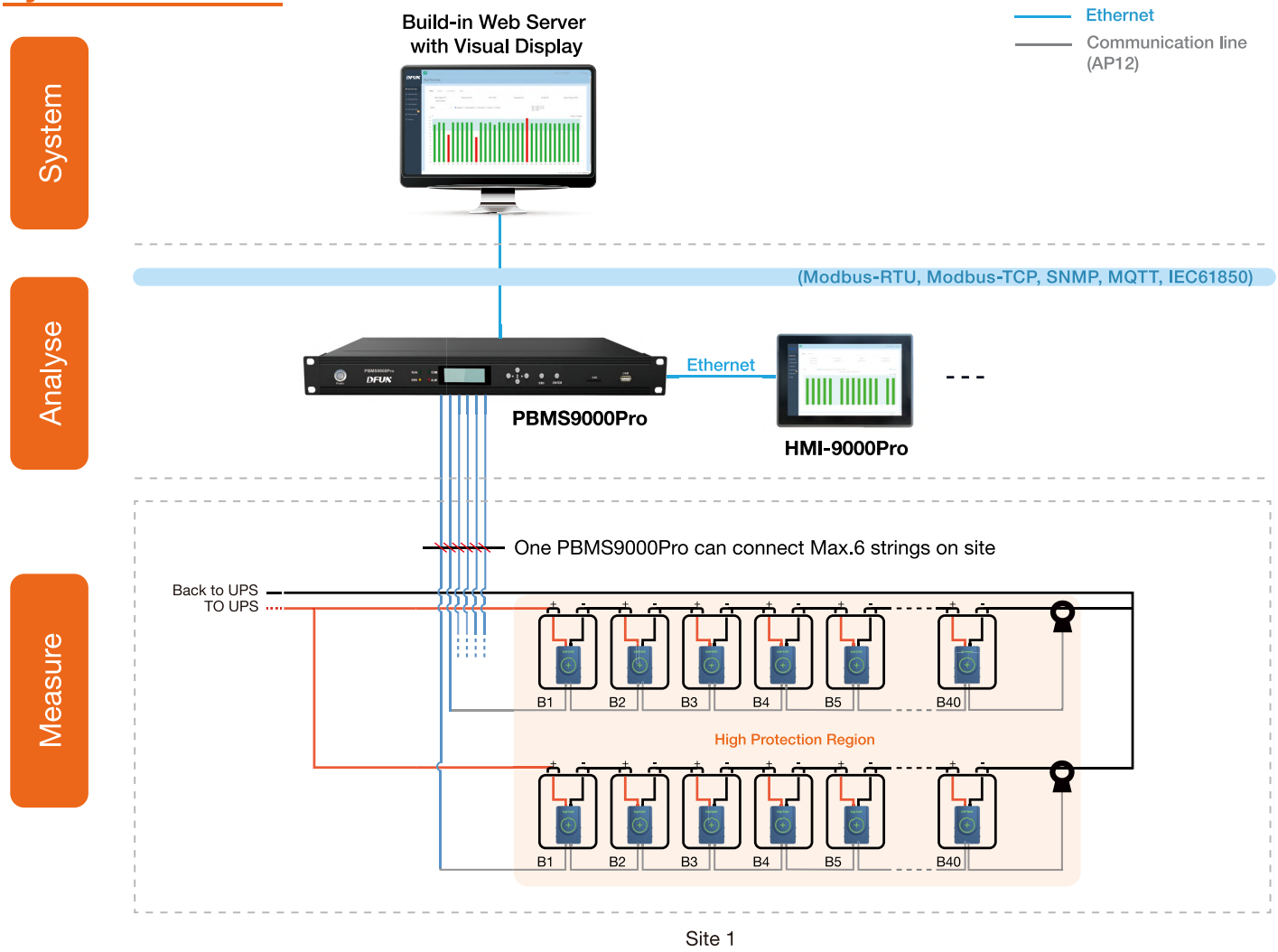
Feature

- Apply to DC power systems like oil & gas, chemical plants, power plants, etc
- Monitor Max. 6 strings and 420pcs batteries in total
- Measure Ni-Cd, Lead-Acid, and multi-pole battery
- Powered by communication bus, no consuming battery power
- Anti-interference design, support to connect with high-frequency UPS
- Auto-sensing for the battery sensor's ID address
- Support Modbus, SNMP, MQTT and IEC61850 protocols
- Support extension sensor for measuring ambient temperature & humidity, leakage current, insulation resistance (optional)
- Comply with IEEE 1188-2005
- Thermal runaway management(Optional temperature and humidity sensor)



PBMS9000Pro for Chemical Plant

System Structure



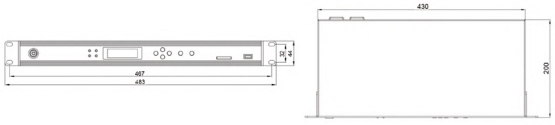
Site 1

Management Layer

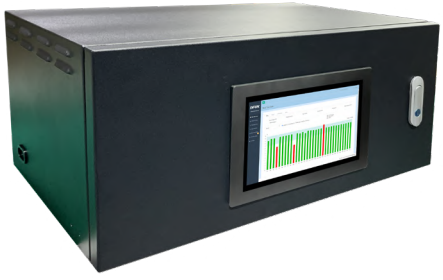
PBMS9000Pro Battery Management Controller

- Standard 1 U design for distribution cabinet
- Built-in web server with visual display
- Monitor Max. 6 strings battery, in a total of 420 batteries
- Measure battery string current & voltage, ripple voltage & current, charge & discharge current, internal temperature, impedance, insulation resistance, leakage current, SOC and SOH
- Setpoint alarm for cell voltage, internal temperature, impedance, SOC, SOH (upper / lower limit) and string voltage, current, SOC (upper / lower limit)
- Alarm notifications via SMS or Email
- Support Modbus-TCP, Modbus-RTU, SNMP and IEC61850 protocols
- Support MQTT for JSON format to data upload
- 1 RS485 port, 2 Ethernet ports and 1 4G Antenna port for data upload
- 6 DI ports (for digital Input connecting)
- 6 DO ports (for sound and light alarms)



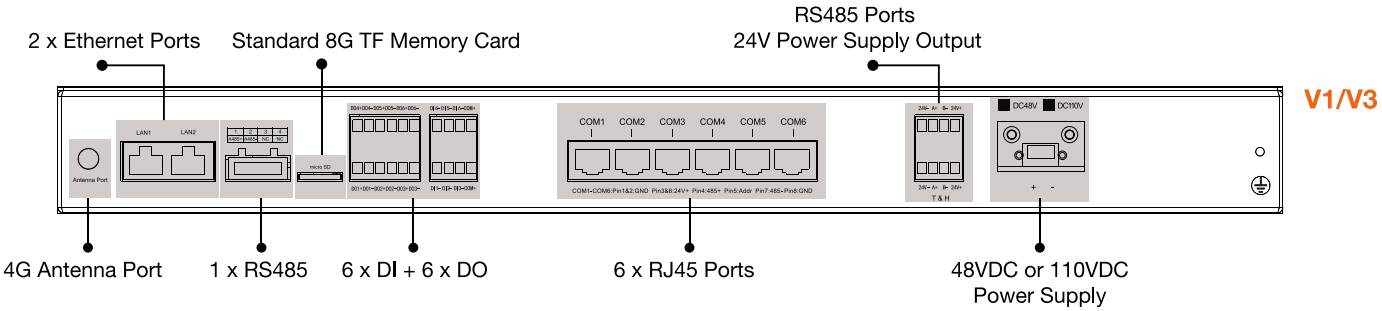


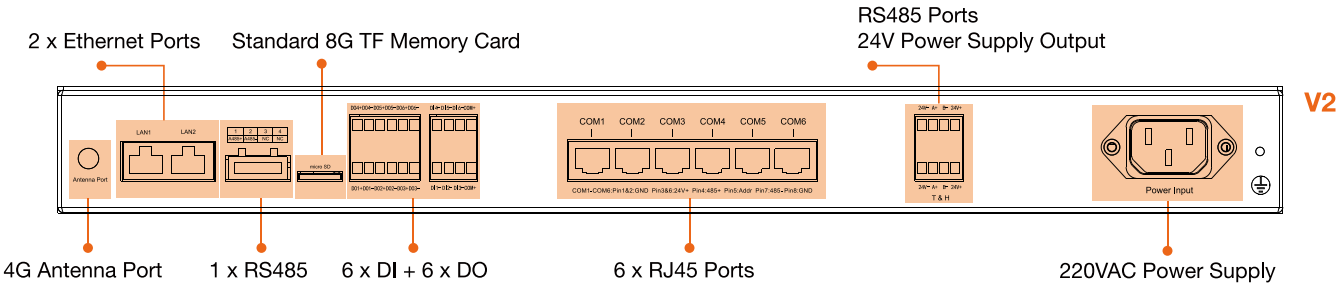
Pre-install & pre-wiring for PBMS9000Pro and HMI-9000Pro (optional)



PBMS9000Pro for Chemical Plant

Interface Description





Technical Specification

CPU	ARM cortex-A7 528MHz	**Up-link Communication**	2 Ethernet ports (10/100M), MODBUS-TCP, SNMP, IEC61850 1 RS485 port, MODBUS-RTU, baudrate: 9600bps, 19200bps, 38400bps (optional)	
Memory	512MB flash, 4G EMMC + 8G TF memory card			
Display	2-inch LCD with backlight	**Down-link Communication**	6 channels RJ45 ports,each port Max. connect 70pcs batteries	
MTBF	≥100,000 hours	**Measure Range**	Voltage	1~6 strings, Range: 20~1000V DC (±0.5%), Resolution:0.01V
Power Supply	V1: 48VDC, Range: 18~72VDC V2: 220VAC, Range: 90~264VAC/127~370VDC V3: 110VDC, Range: 43~160VDC		Current	1~6 strings, Max. Range: -1000~+1000A DC (±1.0%fs,15°C~30°C), Resolution: 0.01A
Dimension	Standard 19-inch 1 U device 483mm × 206mm × 44.5mm (W*D*H) Open hole: 440mm×46mm (L*H)	**Operation Environment**	Operating Temperature: -10~65°C (±1°C) Storage Temperature: -40~70°C Humidity: 10~95% RH, non-condensing	
Power Consumption	<15W (Only main controller)			
Additional Port	6 x DI dry contact 6 x DO relay output, 250VAC/5A or 30VDC/5A	**Extension Sensor (optional)**	-RS485 for Max. 6 strings ambient temperature & humidity measuring -RS485 for 1~6 strings DC leakage current measuring -RS485 for 1~6 strings DC insulation resistance measuring	

String Measuring Sensor

String Current Measuring Sensor - Hall Sensor

- Measure battery string charge and discharge current
- Measure multi-pole battery's string charge and discharge current and ripple current with flexiable module and hall sensor
- Accessories:
  1) Hall Sensor: Range: -1000A~+1000A
  2) Communication cable : 0.3m with AP12 port



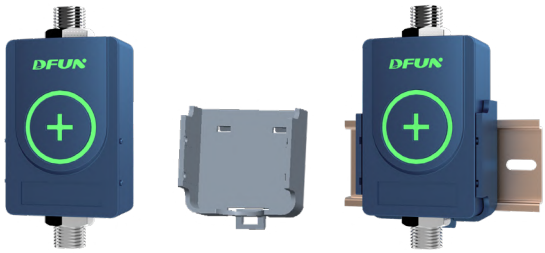
Hall Sensor

Item	Power Supply	Measuring Range	Environment
		String Current	
Hall Sensor	24VDC (Range: 16~32VDC) Power Consumption: <1W	1 Hall Sensor: -1000A~+1000A	Working Temperature: 0°C~45°C Working Temperature Limit: -15°C~55°C

Battery Cell Sensor

PBAT81-02 / PBAT81-12 Battery Cell Sensor

- PBAT81-02 for 2V Lead-Acid battery or 1.2V Ni-Cd battery
  PBAT81-12 for 12V Lead-Acid battery
- Monitor individual battery voltage, internal temperature (negative pole), impedance(ohmic value)
- Auto-balancing
- IP65 protection degree
- UL94-V0 fire rating
- Powered by communication bus, no draw any power from the batteries



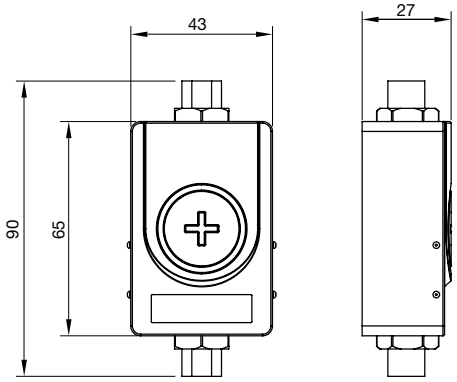
PBAT81 **Base (optional)**

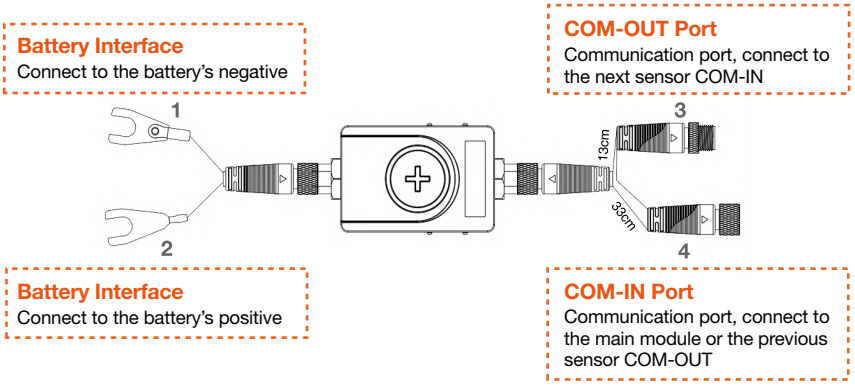
Snap-in DIN-rail Installation

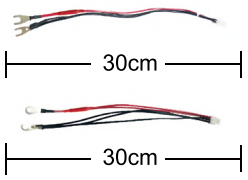
Item	Power Supply	Rated Input Voltage	Measuring Range		
			Voltage	Internal Temperature	Impedance
PBAT81-02	24VDC Power Consumption: < 0.35W	02V / 1.2V	0.5~3VDC (±0.2%)	-20°C ~ 85°C (±0.5°C)	Range: 0.1~50mΩ Repeatability accuracy: 1.0%±25μΩ Consistency accuracy: 1.5%±25μΩ
PBAT81-12		12V	5~18VDC (±0.2%)		

Dimension

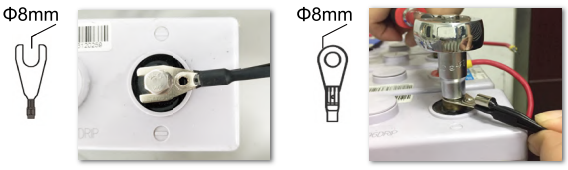
Unit : mm







	Item	Description
Battery Measuring Cable	Length	30cm
	Terminal & Size	U type, hole diameter: 8mm
		O type, hole diameter: 8mm



Order Information

System Structure		Model	Description	Remark
Management Layer	BMS Main Controller	PBMS9000Pro-V1	Battery Management Controller, 48VDC power supply	One per UPS One PBMS9000Pro can monitor Max.6 battery strings (Max.100 batteries)
		PBMS9000Pro-V1-L	Battery Management Controller, 48VDC power supply with 4G	
		PBMS9000Pro-V2	Battery Management Controller, 220VAC power supply	One per UPS One PBMS9000Pro can monitor Max.6 battery strings (Max.420 batteries)
		PBMS9000Pro-V2-L	Battery Management Controller, 220VAC power supply with 4G	
		PBMS9000Pro-V3	Battery Management Controller, 110VDC power supply	One per UPS One PBMS9000Pro can monitor Max.6 battery strings (Max.216 batteries)
		PBMS9000Pro-V3-L	Battery Management Controller, 110VDC power supply with 4G	
	Protocol(optional)	PBMS9000Pro-CP	DFUN standard IEC61850 model available. For customized needs, please contact our sales team.	
Battery Cell Sensor		PBAT81-02-B	02V / 1.2V battery cell sensor	Terminal Material: Metal Aviation Plug
		PBAT81-12-B	12V battery cell sensor	
		PBAT81-DZ	Base, snap-in DIN-rail installation	Each cell sensor includes one DZ (optional)
Accessory	AP to RJ45 Convert Cable	PBAT-COM-AP12-RJ45	Length: 5m	Each string includes one AP12-RJ45
	Cell Sensor Communication Cable	PBAT-COM-AP12-30	IP65 with aviation connectors , length: one side 33cm, other side 13cm	
	Cross-layer Communication cable	PBAT-COM-AP12-500	Cable,Length: 5m	Battery rack cross-layer communication cable, Two AP12(Male and Female)
	Battery Measuring Cable	PBAT-C1/AP12-U-M8	Cable terminal: U type  , hole diameter: 8mm, length: 30cm (standard)	
		PBAT-C1/AP12-O-M8	Cable terminal: O type  , hole diameter: 8mm, length: 30cm	
	Hall Sensor with AP Connector	ST50K2S-M-DF	Rated input: 50A	Measure Range: -50A~+50A, Φ40mm
		ST100K2S-M-DF	Rated input: 100A	Measure Range: -100A~+100A, Φ40mm
		ST200K2S-M-DF	Rated input: 200A	Measure Range: -200A~+200A, Φ40mm
		ST300K2S-M-DF	Rated input: 300A	Measure Range: -300A~+300A, Φ40mm
		ST400K2S-M-DF	Rated input: 400A	Measure Range: -400A~+400A, Φ40mm
		ST500K2S-M-DF	Rated input: 500A	Measure Range: -500A~+500A, Φ40mm
		ST800K2S-M-DF	Rated input: 800A	Measure Range: -800A~+800A, Φ40mm
		ST1000K2S-M-DF	Rated input: 1000A	Measure Range: -1000A~+1000A, Φ40mm
	Group Current Communication Wire	PBAT-COM-AP12-HALL-200	Length: 2m, both AP12 female connectors,5-pin female to PBAT81, 6-pin female to Hall	
Optional Module	HMI for Local Display & Operation	HMI-9000Pro	10-inch touch-screen HMI for local display & operation	Power Supply from Auxiliary Power Module
	Auxiliary Power Module (choose one out of three)	HDR-30-12	220V AC to 12V DC power supply	Provide power supply for the HMI-9000Pro
		DDR-15L-12	48V DC to 12V DC power supply	
		RSD-30H-12	110V DC to 12V DC power supply	
	Cabinet for Pre-Wiring in Factory	PBMS-BOX-260	Dimension: 535*260*335mm (W*H*D)	Pre-install & pre-wiring for PBMS9000Pro and HMI-9000Pro
		PBAT-COM-AP12-1000	Cable,Length: 10m	Cabinet to first cell Cable,Two AP12 (Male and Female)
		PBAT-COM-AP12-1500	Cable,Length: 15m	

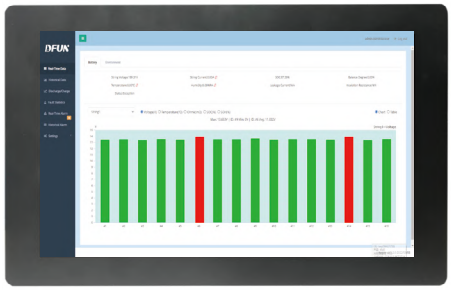
Note: If battery terminal is M5, M6, M10, M12, M16, please contact our sales for customize requirements before placing the order.

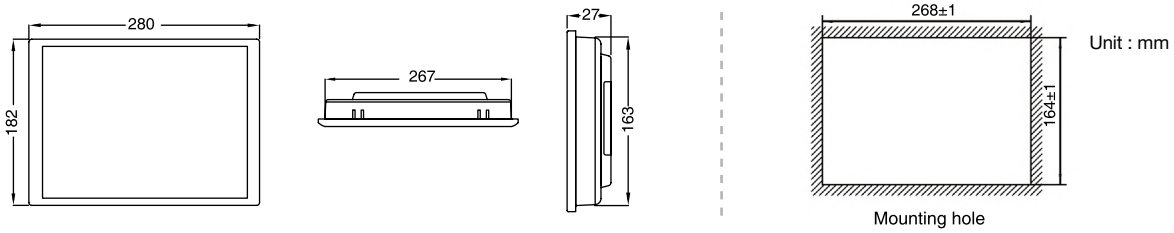
HMI (Optional)

Local Display & Operation for Different Main Controller

10.1-inch Android HMI

- 10.1 -inch touch-screen HMI for local display and operation
- OS: Android 10.0
- CPU: R818 (4-core A53 1.6GHz)
- Running memory: 1GB,
- EMMC: 8GB
- Support the reading and configuration of all parameters of the system





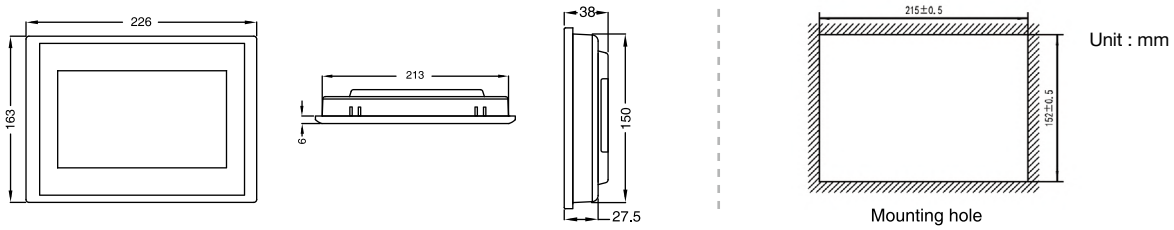
Order Information

Model	Description	Remark
HMI-2000Pro	One per PBMS2000Pro main controller	One per PBMS2000Pro, communication with PBMS2000Pro via Ethernet
HMI-9000	One per PBMS9000 main controller	One per 12 (max) PBMS9000, Communication with PBMS9000 via Ethernet
HMI-9000Pro	One per PBMS9000Pro main controller	One per 12 (max) PBMS9000Pro, Communication with PBMS9000Pro via Ethernet

7-inch Normal HMI

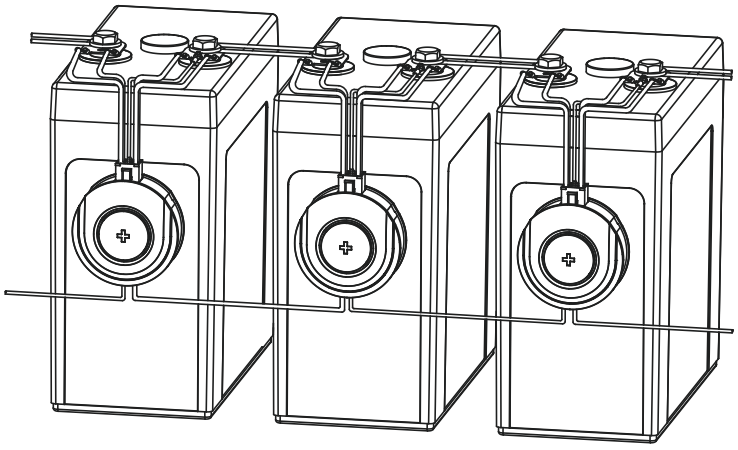
- 7-inch touch-screen HMI for display and operation
- **Real-time Data Inquiry:**
  1) Individual battery voltage, impedance, inner temperature, SOC, SOH
  2) String voltage, charge and discharge current, SOC, balance degree
- **Alarm Data Inquiry:**
  1) Real-time alarm status and related alarm information (faulty battery ID, date/time, alarm reason, communication status, etc.)
  2) 3000 alarm records for each string battery





Order Information

Model	Description	Remark
HMI-2000	One per PBMS2000 main controller	One per PBMS2000, communication with PBMS2000 via RS485



Case Study & Our Clients

Chapter 3

Case Study

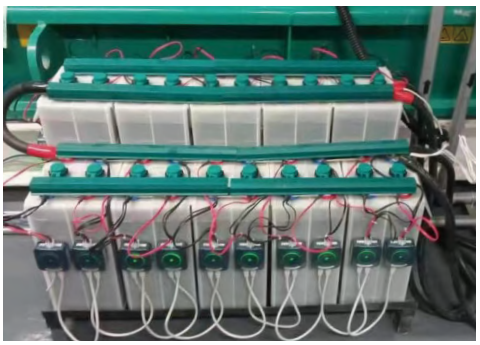
Data Center

Thailand True IDC Bangna Data Center
Thailand Suvarnabhumi Airport
Malaysia Intel Data Center
Malaysia Ekova Data Center
Qatar OOREDOO Google Data Center
Qatar Lusail Towers Data Center
Dubai Gulf Data Hub Data Center
Chile Entel Data Center
Chile Claro Telecommunication
Singapore Changi Airport
Indonesia TELKOMSEL Server Room
Indonesia PLN Island Energy Storage
South Africa MTN Data Center
Turkey Turkcell Date Center
Vietnam VIETEL Data Center
ChinDATA - ChinaNorth H02 Data Center
.....



Oil & Gas

BASF Chemical Factory International
Saudi Arabian Oil Campany (Saudi Aramco)
Malaysia Petronas
Morocco O.C.P Phosphate Factory



Transportation

Russia Moscow Metro Station
Australia Northconnex Traffic
Sydney North Connex Rail
Thailand MRT Yellow line & Pink Line
XinJiang RailWay Station (Line 1)
HeFei RailWay Station (Line 2)
Chengdu Modern Tram (Line 2)
Changsha Metro Station (Line 4)
Harbin RailWay Station (Line 3)
.....



Bank & Substation

CHINA CITIC BANK (30 Branches)
Malaysia Bank of Negara
Thailand KTB BANK
Thailand GHB (Government Housing Bank)
Iran HREC Substation
Romania TRANSELECTRICA Substation
Romania Nama Raureni Substation
Singapore Panitan-Nanbas Substation Line 2
Greece IPTO Substation
.....



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Telkom
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Gulf Data Hub


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GROUP


intel


LUMEN


MTR港鐵


socomec
Innovative Power Solutions


Schneider
Electric


DELTA


riello ups


EATON


VERTIV


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Saudi Aramco


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With Advanced Technology And
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Your Representative